

Basics of University Mathematics

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LUT School of Engineering Sciences

Week 1
September 1–2, 2025

Practical Matters

- **Responsible teacher:** Vesa Kaarnioja (vesa.kaarnioja@lut.fi).
- All course material will be available on the course's Moodle page.
- To complete the course, you must:
 1. Pass the course exam **and**
 2. Complete the mathematics exercise quota (earning a cumulative 30% of mathematics exercises points)[†], **and**
 3. Complete the MATLAB exercise quota (20 MATLAB assignments successfully solved).
- The course consists of weekly lectures (3h+3h), mathematics exercise groups (2h), and MATLAB exercise groups (2h).
- Please register for the exercise groups on a weekly basis on the course Moodle page (under the “Week 1”, “Week 2”, “Week 3”, . . . , tabs).
- Mathematics exercises start already the first week, MATLAB exercises start on the second week.

[†]If you have already completed the MATLAB quota in 2024 or earlier, you can directly attend the exam without retaking the mathematics/MATLAB exercises.

Lectures

Monday, September 1, 14.15–17.00 Auditorium 2310	Tuesday, September 2, 8.30–11.00 Viipuri hall
Monday, September 8 No lecture	Tuesday, September 9, 8.30–14.00 Student Union building, auditorium 103
Monday, September 15, 14.15–17.00 Auditorium 2310	Tuesday, September 16, 8.30–11.00 Viipuri hall
Monday, September 22 No lecture	Tuesday, September 23, 8.30–14.00 Viipuri hall (8.30–11.00), Student Union building, auditorium 103 (11.15–14.00)
Monday, September 29, 13.15–16.00 Student Union building, auditorium 103	Tuesday, September 30, 13.15–16.00 Student Union building, auditorium 103
Monday, October 6 No lecture	Tuesday, October 7, 8.30–11.00, 13.15–16.00 Viipuri hall (8.30–11.00), Auditorium 2310 (13.15–16.00)
Monday, October 13, 14.15–17.00 Auditorium 2310	Tuesday, October 14, 10.15–13.00 Auditorium 2310

Exercise Practices

- This course uses an **exercise quota**: in order to pass the course, students must complete a certain number of exercises (including *both* mathematics exercises *and* MATLAB exercises). You may obtain extra points for the exam by completing additional mathematics exercises beyond the required quota.
 - **Week 1: Basic Skills Test.** Register for any of the course's exercise groups and attend the session (please bring your own laptop). The basic skills test will be conducted during the exercise session under supervision. Completing the test will earn you extra points toward the course grade.
 - **Weeks 2–7:** The weekly exercise problems must be solved before the exercise sessions. At the beginning of each session, a name list is passed around, and each student marks the problems they are prepared to present in front of the class. The instructor will then choose students from the list to present their solutions. Points are awarded only for the problems marked on the list.

Exercise Points

Completing more than 30% of the course's exercises accrues bonus points, which can be used to improve your exam grade according to the following table:

At least 30% of exercise points	completion requirement
At least 40% of exercise points	1 course exam point
At least 50% of exercise points	2 course exam points
At least 60% of exercise points	3 course exam points
At least 70% of exercise points	4 course exam points
At least 80% of exercise points	5 course exam points
At least 90% of exercise points	6 course exam points

(6 course exam points corresponds to one perfectly solved exam question.)

The exercise points are valid for the exams on 24 October 2025, 4 December 2025, and 28 January 2026.

Support for Exercises

- We will organize a voluntary “tutoring lab” for this course on Tuesdays at 17-19 (room 3305) and Thursdays at 8-10 (room 3307). You can attend these sessions to solve the course exercises collaboratively with the support of the course assistants.
 - The Tuesday tutoring lab will also serve as a weekly Question time for this course. You do not need to register for the tutoring labs.
- In addition, there is a second “tutoring lab” (“laskaripaja”) organized by PhD students at the Department of Computational Engineering, who are happy to assist with this course’s assignments:
 - **Time:** Mondays and Wednesdays at 16-18 in room Yo-talo 216 (near the guild room).
 - **First meeting:** Monday September 8.

Schedule

Week 1: Register for one of this week's exercise groups by **Tuesday 8pm (tomorrow)**. You do not need to solve any problems in advance, simply attend the exercise group you registered to and bring your own laptop.

Weeks 2–7: Register to one MATLAB exercise group and one mathematics exercise group each week. The registration for each week's exercise groups closes on **Tuesday 8pm**.

Set Theory

Set Theory

This chapter introduces the concept of a set as well as other related concepts and operations. These belong to a branch of mathematics called **set theory**. It is a very fundamental part of mathematics that lays the foundation for precisely discussing more complex topics. In practice, this chapter introduces notations that will be needed in this and subsequent courses.

Sets

In mathematics, there is often a need to talk about collections formed by multiple numbers. These are called **sets** of numbers, and the numbers belonging to them are called **elements** (of the set). For example, the formula

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

applies to every positive integer n , but not to other types of numbers. This can be succinctly expressed by adding the notation $n \in \mathbb{Z}_{>0}$ at the end of the formula, which is read precisely as “ n belongs to the set $\mathbb{Z}_{>0}$ ” and practically means that n can be any positive integer. Similarly, the notation $n \notin \mathbb{Z}_{>0}$ means that n does not belong to the set $\mathbb{Z}_{>0}$.

Some important sets have their own symbols, like the integers mentioned above. The most important ones are:

- **Natural numbers** $\mathbb{N} = \{0, 1, 2, 3, \dots\}$
- **Integers** $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$
- **Rational numbers** (i.e., repeating decimal numbers)
 $\mathbb{Q} = \left\{ \frac{m}{n} \mid m, n \in \mathbb{Z}, n \neq 0 \right\}$
- **Irrational numbers** (i.e., non-repeating decimal numbers)
- **Real numbers** (i.e., all rational and irrational numbers together) $\mathbb{R}$
- **Complex numbers** $\mathbb{C} = \{ z \mid z = x + yi, x, y \in \mathbb{R}, i^2 = -1 \}$
(not covered in this course)

The notations $\mathbb{N}$, $\mathbb{R}$, and $\mathbb{C}$ are intuitive (**n**atural, **r**real, **c**omplex). The origin of the symbol $\mathbb{Z}$ is the German word **Z**ahlen, which means number. The notation for rational numbers comes from the Italian word **q**uoziente, meaning quotient, which is similar to the corresponding English word.

- Positive real numbers $\mathbb{R}_{>0}$, $\mathbb{R}_+$, or $\mathbb{R}^+$ (similarly for sets $\mathbb{Z}$ and $\mathbb{Q}$)
- Non-negative real numbers $\mathbb{R}_{\geq 0}$ or $\mathbb{R}_0^+$ (similarly for sets $\mathbb{Z}$ and $\mathbb{Q}$)
- Negative real numbers $\mathbb{R}_{<0}$, $\mathbb{R}_-$, or $\mathbb{R}^-$ (similarly for sets $\mathbb{Z}$ and $\mathbb{Q}$)
- Non-positive real numbers $\mathbb{R}_{\leq 0}$ or $\mathbb{R}_0^-$ (similarly for sets $\mathbb{Z}$ and $\mathbb{Q}$)
- Non-zero real numbers $\mathbb{R}_{\neq 0}$ (similarly for sets $\mathbb{Z}$ and $\mathbb{Q}$).

The capital letters used as names for the sets in the previous list are reserved for those specific sets. Other sets are typically denoted by ordinary capital letters A , B , C , etc.

On the other hand, as seen from the previous list, the elements of sets are presented within curly braces. If there are only a few elements, they can all be listed. For example, $A = \{1, 2, 3, 5, 8\}$ is a sensible way to denote the set formed by the numbers 1, 2, 3, 5, and 8. If there are infinitely many numbers but they follow a regular enough pattern, the elements of the set can still be listed using the notation[†] $\dots$, as in the case of natural and integer numbers.

[†]The notation $a_1, a_2, a_3, \dots$ means that the list continues indefinitely (according to the rule deducible from the given numbers a_1, a_2, a_3). The notation $a_1, a_2, a_3, \dots, a_n$ means the same, except that the list now ends with the number a_n . The interpretation is the same for the notations $a_1 + a_2 + \dots$ and $a_1 + a_2 + \dots + a_n$ and other similar cases.

Often, however, sets are so irregular that their elements cannot be listed. For this, a more general set notation is needed, which has already been used above to describe rational numbers. In general, the notation takes the form

$$A = \{ \text{arbitrary element type} \mid \text{conditions required for the arbitrary element} \} \quad (1)$$

and is called **set-builder notation**. If multiple conditions need to be met for the elements, they are listed separated by “and” or commas. If it is sufficient that at least one of several conditions is met, the conditions are separated by “or.”

Example

The following notations mean the same thing:

$$\begin{aligned}\{x \in \mathbb{R} \mid 0 < x < 1\} &= \{x \in \mathbb{R} \mid x > 0 \text{ and } x < 1\} \\ &= \{x \in \mathbb{R} \mid x > 0, x < 1\},\end{aligned}$$

which refers to the set containing all real numbers greater than zero and less than one.

The set of real numbers whose distance from the origin is greater than 1 can be written as $\{x \in \mathbb{R} \mid x > 1 \text{ or } x < -1\}$.

The type of elements (real number, rational number, ...) is usually indicated before the vertical bar, but it can also be included among the conditions. A colon can also be used instead of a vertical bar. The following notations, therefore, mean the same set:

$$\{x \in \mathbb{Q} \mid x > 1\} = \{x \mid x \in \mathbb{Q} \text{ and } x > 1\} = \{x : x \in \mathbb{Q} \text{ and } x > 1\}.$$

Basic Exercise

Write the set consisting of all integers whose squares are greater than 1 but less than 50 using the notation (1).

The elements of sets do not actually have to be numbers; they can be any type of similar elements, such as points, vectors, functions, or sets. For example, the circumference of a circle with center at the origin and radius 2 can be expressed as a set of points:

$$\{(x, y) \mid x \in \mathbb{R}, y \in \mathbb{R}, x^2 + y^2 = 4\} = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 4\}.$$

Intervals

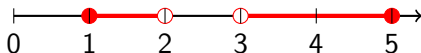
Functions are often considered over all values between two real numbers. For example, the function $f(x) = \sqrt{1 - x^2}$ is defined only for those values of the variable x that lie between the numbers -1 and 1 . Such sets of numbers are called **intervals**. Intervals are categorized as closed, open, and half-open depending on whether the endpoints of the interval are included or not. Below is a summary of different types of intervals:

- closed interval $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$
- left half-open interval $(a, b] = \{x \in \mathbb{R} \mid a < x \leq b\}$
- right half-open interval $[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\}$
- open interval $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$.
- unbounded above interval $(a, \infty) = \{x \in \mathbb{R} \mid x > a\}$
- unbounded below interval $(-\infty, b) = \{x \in \mathbb{R} \mid x < b\}$.

Note that brackets are not used with infinity. This is because infinity is not a number, so equality cannot be allowed at the endpoint

In the case of an open or half-open interval, a reversed bracket can also be used instead of a parenthesis. For example, $(3, 6]$ can also be written as $]3, 6]$.

The accompanying figure shows the half-open intervals $[1, 2)$ and $(3, 5]$.



Example

- Open intervals[†] $(-1, 2)$, $(3, 9)$.
- Half-open intervals: $[2, 4)$, $(4, 5]$.
- Closed intervals: $[4, 7]$, $[-10, 8]$.

Basic Exercise

- Write the set $\{x \in \mathbb{R} \mid 3 < x \leq 7\}$ as an interval.
- Write the interval $[-2, 4)$ using the notation (1) .
- Write the set $\mathbb{R}_{<0}$ as an interval.

[†]Incidentally, note that the interval $(-\infty, a)$ is open, the interval $(-\infty, a]$ is closed, and the interval $(-\infty, \infty) = \mathbb{R}$ is both open and closed. A more detailed explanation is omitted, as it would require a precise definition of open and closed sets, which is not relevant to this course.

Subsets

Every rational number is an element of the set of real numbers. Specifically, it is said that the rational numbers are a **subset** of the real numbers, denoted by $\mathbb{Q} \subset \mathbb{R}$.

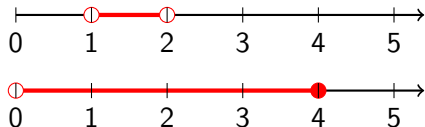
If every element of set A belongs to set B , i.e., the following holds:

$$x \in A \implies x \in B,$$

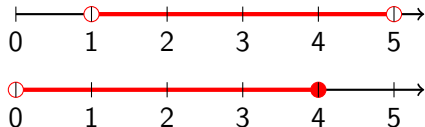
then set A is called a subset of set B , and it is denoted as $A \subset B$.

Example

- (a) The set $(1, 2)$ is a subset of the set $(0, 4]$, because all elements of the former set are included in the latter. This is evident from the figure.



- (b) The set $(1, 5)$ is not a subset of the set $(0, 4]$, because not all elements of the former set are included in the latter.



Example (continued)

(c) We have

$$\{1, 2, 4\} \subset \mathbb{N},$$

but

$$\{-1, 2, 4\} \not\subset \mathbb{N} \quad \text{and} \quad \{1, \tfrac{3}{2}, 4\} \not\subset \mathbb{N}.$$

(d)

$$\{x \in \mathbb{R} \mid x = 1 \text{ or } 2 < x < 5\} \subset \mathbb{R}$$

(e) For all $a, b \in \mathbb{R}$ it holds that

$$[a, b] \subset \mathbb{R},$$

but not

$$[a, b] \not\subset \mathbb{Q}.$$

The latter only holds in the special case $b = a$.

(f)

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

Example

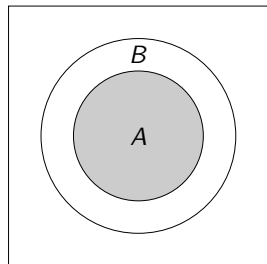
The notation $f: \mathbb{R}_{\neq 0} \rightarrow \mathbb{R}$ (which is discussed further in Chapter 1.3.8) means that we are considering functions named f whose domain is $\mathbb{R}_{\neq 0}$.

The notation $f: A \rightarrow \mathbb{R}$, $A \subset \mathbb{R}_{\neq 0}$, means that we are considering functions f whose domain A does not include zero. Thus, A can be, for example, $[1, 2]$, unlike in the previous case where the domain could only be $\mathbb{R}_{\neq 0}$. Hence, $A \subset \mathbb{R}_{\neq 0}$ is a short way to express all different sets composed of real numbers that do not include zero.

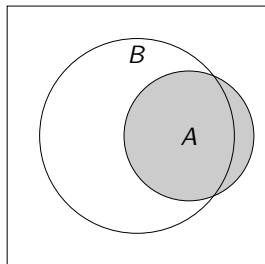
If we want to express that set A is not a subset of set B , we can denote it as $A \not\subset B$.

Venn Diagrams

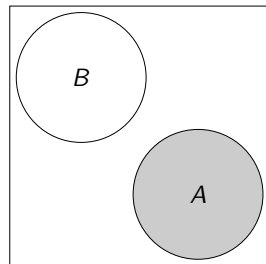
Some concepts related to sets of numbers can be conveniently illustrated with images where each set corresponds to a certain area. For example, a subset of set B corresponds to an area that is entirely contained within the area representing set B , as shown in the following images.



$$A \subset B$$



$$A \not\subset B$$



$$A \not\subset B$$

Images like the ones above are called **Venn diagrams**, and they can illustrate set theory concepts regardless of the type of elements in the sets.

Basic Exercise

Determine whether the following statements are true based on Venn diagrams.

(a) If $A \subset B$ and $B \subset C$, then
 $A \subset C$.

(b) If $A \subset B$ and $B \subset A$, then
 $A = B$.

Set Operations

We can define operations between sets as well:

- Union: $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
- Intersection: $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$
- Difference: $A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}$.

To reiterate, in the expression “ $x \in A$ and $x \in B$ ”, the word[†] **and** means that both conditions $x \in A$ and $x \in B$ must be satisfied simultaneously, in accordance with common language usage. Similarly, in cases with multiple conditions, all of them must be fulfilled at the same time. Therefore, for example, the set $\mathbb{R} \cap \mathbb{R}_{>0}$ simplifies to the set $\mathbb{R}_{>0}$, because

$$\begin{aligned}\mathbb{R} \cap \mathbb{R}_{>0} &= \{x \mid x \in \mathbb{R} \text{ and } x \in \mathbb{R}_{>0}\} = \{x \mid x \in \mathbb{R} \text{ and } x > 0\} \\ &= \{x \mid x > 0\} = \mathbb{R}_{>0}.\end{aligned}$$

[†]More precisely, the words **and** and **or** are logical connectives, i.e., logical operations that combine logical propositions, which are elements of logic.

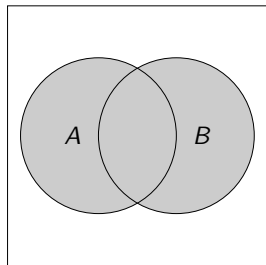
On the other hand, in the expression “ $x \in A$ or $x \in B$ ”, the word **or** means, contrary to common language, that at least one of the conditions $x \in A$ or $x \in B$ must be satisfied, or **both**. In cases with more than two conditions, it suffices that any one of them is fulfilled. Thus, for example, it holds that $\mathbb{R}_{<0} \cup \{0\} \cup \mathbb{R}_{>0} = \mathbb{R}$ (Write out the intermediate steps). If conditions are separated by a comma, the word “and” is implied.

Example

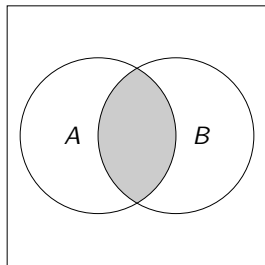
Non-negative real numbers can also be denoted as $\mathbb{R}_+ \cup \{0\}$ and $\mathbb{R} \setminus \mathbb{R}_-$. How can you similarly denote real numbers excluding zero?

Care should be taken with the use of “and” or “or” in this course, especially when solving inequalities. For example, if both intervals $(1, 2)$ and $[3, 4]$ are solutions, then the answer should be written as $x \in (1, 2)$ or $x \in [3, 4]$, and not $x \in (1, 2)$ and $x \in [3, 4]$, as the latter conditions cannot be fulfilled simultaneously, implying no solutions exist.

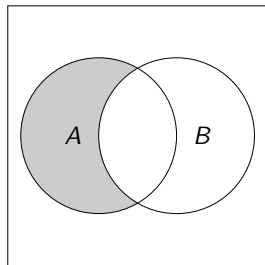
The interpretations of set operations with number sets are clearly illustrated in the accompanying Venn diagrams.



$$A \cup B$$



$$A \cap B$$



$$A \setminus B$$

Example

- Unions: $[4, 6] \cup [5, 7) = [4, 7)$, $\{7\} \cup [5, 7) = [5, 7]$.
- Intersections: $[4, 6] \cap [5, 7) = [5, 6]$, $[4, 6) \cap [5, 7) = [5, 6)$, $\mathbb{N} \cap [3.4, 6) = \{4, 5\}$.
- Differences: $[4, 6] \setminus \{6\} = [4, 6)$, $[4, 6] \setminus (5, 8] = [4, 5]$.

Example

(a) As an example, let's write a set in interval notation according to (1):

$$\begin{aligned} & \{x \in \mathbb{R} \mid -1 < x < 1 \text{ or } 2 < x \leq 3\} \\ &= \{x \in \mathbb{R} \mid -1 < x < 1\} \cup \{x \in \mathbb{R} \mid 2 < x \leq 3\} \\ &= (-1, 1) \cup (2, 3]. \end{aligned}$$

Often, the second intermediate step is omitted.

Example (continued)

(b) Let's write a set expressed in intervals using the notation (1):

$$\begin{aligned}(1, 3) \cap (2, 4) &= \{x \in \mathbb{R} \mid 1 < x < 3\} \cap \{x \in \mathbb{R} \mid 2 < x < 4\} \\ &= \{x \in \mathbb{R} \mid 1 < x < 3 \text{ and } 2 < x < 4\} \\ &= \{x \in \mathbb{R} \mid 2 < x < 3\}.\end{aligned}$$

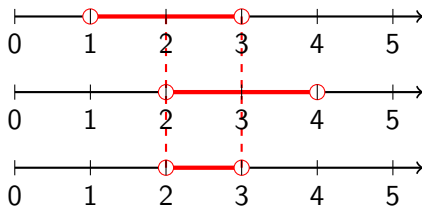
Often, the second intermediate step is omitted. Of course, one could also write

$$(1, 3) \cap (2, 4) = (2, 3) = \{x \in \mathbb{R} \mid 2 < x < 3\}.$$

(c) Let's take an example of a situation where set difference is typically used:

$$\{x \in \mathbb{R} \mid x^2 - 3x + 2 \neq 0\} = \{x \in \mathbb{R} \mid x \neq 1 \text{ and } x \neq 2\} = \mathbb{R} \setminus \{1, 2\}.$$

The interval to which the union or intersection of intervals simplifies can be easily inferred from a diagram like the one corresponding to part (b) of the previous exercise.



Basic Exercise

(a) Write the following sets using interval notation:

(i) $\{x \in \mathbb{R} \mid x > 2 \text{ and } x \leq 99\}$

(ii) $\{x \in \mathbb{R} \mid 1 < x \leq 3 \text{ or } 2 \leq x \leq 6\}$

(iii) $\{x \in \mathbb{R} \mid 1 < x \leq 3 \text{ and } 2 \leq x \leq 6\}$

(iv) $\{x \in \mathbb{R} \mid x^2 > 9, x > 0\}$

(v) $\{x \in \mathbb{Q} \mid x^2 > 9, x > 0\}$.

(b) Write the following sets using set-builder notation:

(i) $[-2, 3)$

(ii) $(-10, -1) \cup [0, 2]$

(iii) $(-\infty, 6]$.

It should also be noted that the **empty set** is the concept in set theory corresponding to zero. It refers to a set that contains no elements, and is denoted by $\emptyset$ or $\{\}$. For the empty set, for example, the property $A \cup \emptyset = A$ holds, similar to how $a + 0 = a$ holds for zero.

In basic courses, set-theoretic operations primarily come into play when defining the domains of functions. For example, in probability theory, they are also used in calculations when forming the set of favorable outcomes.

All sets contain the empty set as a subset. For example, $\emptyset \subset \{1, 2, 4\}$, $\emptyset \subset (1, 2]$, $\emptyset \subset \mathbb{R}$.

Sum and Product Notations $\sum$ and $\prod$

In mathematics, we sometimes encounter long sums such as $1^2 + 2^2 + 3^2 + \cdots + 15^2$. Such sums can be compactly represented using the summation notation $\sum$.[†] For example, we have

$$\sum_{i=1}^{15} i^2 = 1^2 + 2^2 + 3^2 + \cdots + 15^2. \quad (2)$$

The expression i^2 is called the **general term** of the sum, and its variable i is the **index of summation**. Thus, in the previous case, the index of summation iterates through all integers from the lower bound 1 to the upper bound 15.

[†] $\sum$ is the uppercase Greek letter sigma, which corresponds to the first letter of the word sum.

Example

Write the sum $1 + 4 + 27 + 256 + 3125$ using summation notation.

Solution

We need to find an expression dependent on the integer i such that the terms of the sum are obtained by successive values of i . The terms of the sum seem to grow rapidly, indicating that the general term of the sum is likely a power or exponential function. With this in mind, and by trying to find something systematic in the sum, we notice that the second and third terms are familiar powers of integers: $4 = 2^2$, $27 = 3^3$. It can be verified that this pattern also holds for the other terms in the sum. Thus, the terms of the sum are of the form i^i , where $i = 1, 2, 3, 4, 5$. Therefore, we have

$$1 + 4 + 27 + 256 + 3125 = \sum_{i=1}^5 i^i.$$

Verification:

$$\sum_{i=1}^5 i^i = 1^1 + 2^2 + 3^3 + 4^4 + 5^5 = 1 + 4 + 27 + 256 + 3125.$$

In the above, both sums start with the index 1, but this is not mandatory. For example, sum (2) can also be written as

$$\begin{aligned}\sum_{i=0}^{14} (i+1)^2 &= (0+1)^2 + (1+1)^2 + (2+1)^2 + \cdots + (14+1)^2 \\ &= 1^2 + 2^2 + 3^2 + \cdots + 15^2.\end{aligned}$$

For an arbitrary sum, we use the notation

$$\sum_{i=m}^n a_i = a_m + a_{m+1} + \cdots + a_{n-1} + a_n.$$

Summations satisfy the following rules:

$$(a) \sum_{i=m}^n (ca_i) = c \sum_{i=m}^n a_i$$

$$(b) \sum_{i=m}^n (a_i \pm b_i) = \sum_{i=m}^n a_i \pm \sum_{i=m}^n b_i,$$

which can be easily justified by expanding the sums. For example, it holds that

$$\sum_{i=m}^n (ca_i) = ca_m + ca_{m+1} + \cdots + ca_n = c(a_m + a_{m+1} + \cdots + a_n) = c \sum_{i=m}^n a_i.$$

Often, formulas are written using summation notation to achieve a more compact form. For instance, compare the Taylor polynomial notations

$$P_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n$$

and

$$P_n(x) = \sum_{i=0}^n \frac{f^{(i)}(a)}{i!}(x - a)^i.$$

However, this does not mean that the formulas are in their most illustrative form. **If handling an expression using summation notation seems less clear, it is advisable to expand it, work with it in this form, and then write it back as a sum**, as done in the previous proof.

A sum $a_m + a_{m+1} + \cdots$, which extends indefinitely unlike the previous cases, is called a **series**, and it is denoted by

$$\sum_{i=m}^{\infty} a_i = a_m + a_{m+1} + \cdots .$$

Basic Exercise

(a) Expand the sum

$$\sum_{i=2}^6 (-1)^i \frac{2i-1}{2i}.$$

(b) Write

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$$

using summation notation.

(c) Write the normal form of a higher-degree polynomial equation

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$$

more concisely using summation notation.

In part (a), the factor $(-1)^i$ causes the sign of the terms in the sum to alternate. Such a sum is called **alternating**.

Double summation

Sometimes, double summations are also needed. Let's explore this with an example.

Example

Consider the array[†]

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

- (a) Write the sum of the elements of the matrix using double summation notation.
- (b) Write the sum of the diagonal elements from the top left to the bottom right, and the elements below these diagonals, using double summation notation.

[†]Such mathematical objects are later referred to as **matrices** in Chapter 4.

Solution

- (a) First, observe that the sum of the elements can be written more compactly as

$$a_{11} + a_{12} + a_{13} + a_{21} + a_{22} + a_{23} + a_{31} + a_{32} + a_{33} = \sum_{j=1}^3 a_{1j} + \sum_{j=1}^3 a_{2j} + \sum_{j=1}^3 a_{3j}.$$

(What is actually being calculated with each sum?) Now, the three similar sums can be combined into one double sum:

$$\sum_{j=1}^3 a_{1j} + \sum_{j=1}^3 a_{2j} + \sum_{j=1}^3 a_{3j} \stackrel{(1)}{=} \sum_{i=1}^3 \left(\sum_{j=1}^3 a_{ij} \right) = \sum_{i=1}^3 \sum_{j=1}^3 a_{ij},$$

where the notation $\sum_{i=1}^3 \sum_{j=1}^3 a_{ij}$ is just a shorter way to denote this double sum. The notation $\sum_{j=1}^3 a_{ij} = a_{i1} + a_{i2} + a_{i3}$ represents the sum of the elements of an arbitrary row i .

Solution (continued)

- (b) In the previous part, the sum of the elements of each row was written as a separate sum. Doing the same now, we see that:

$$a_{11} = \sum_{j=1}^1 a_{1j}$$

$$a_{21} + a_{22} = \sum_{j=1}^2 a_{2j}$$

$$a_{31} + a_{32} + a_{33} = \sum_{j=1}^3 a_{3j}.$$

These similar sums can again be combined using another sum:

$$\sum_{j=1}^1 a_{1j} + \sum_{j=1}^2 a_{2j} + \sum_{j=1}^3 a_{3j} = \sum_{i=1}^3 \sum_{j=1}^i a_{ij},$$

where $\sum_{j=1}^i a_{ij}$ represents the sum of the elements in an arbitrary row i .

Basic Exercise

- (a) Expand the double sum

$$\sum_{m=2}^4 \sum_{n=1}^m mn.$$

- (b) Write the sum of the elements of the matrix from the previous exercise as a double sum where the inner sum represents the sum of the elements in each column. Verify your answer by expanding it.

Exercise

Write the sum of the diagonal elements and those above them from the matrix in the previous exercise using double summation notation by summing (a) the elements of the columns (b) the elements of the rows in the inner sum. Verify your answers by expanding them.

Factorial

Before discussing product notation, let us review the concept of **factorial**, denoted by

$$n! = 1 \cdot 2 \cdot \dots \cdot (n-1) \cdot n,$$

where n is a non-negative integer. Note that, perhaps counterintuitively, $0! = 1$ holds. This convention for the factorial of zero allows formulas to work more smoothly. For example, the previously mentioned Taylor polynomial can be written in a unified form

$$P_n(x) = f(a) + \sum_{i=1}^n \frac{f^{(i)}(a)}{i!} (x-a)^i$$

instead of

$$P_n(x) = \sum_{i=0}^n \frac{f^{(i)}(a)}{i!} (x-a)^i,$$

where we also use the convention $f^{(0)}(a) = f(a)$. The factorial of zero is discussed in more detail in Appendix 6.3.

Product

The regular product $1^2 \cdot 2^2 \cdot 3^2 \cdot \dots \cdot 15^2$ can be written more compactly as

$$\prod_{i=1}^{15} i^2 = 1^2 \cdot 2^2 \cdot 3^2 \cdot \dots \cdot 15^2.$$

The product notation $\prod$ is used similarly to the summation notation.[†]

Basic Exercise

(a) Write out the product

$$\prod_{j=1}^{20} (2j - 1).$$

(b) Perform the division

$$\frac{16!}{9!}$$

and write the resulting expression using product notation.

[†] $\prod$ is a capital pi, the Greek letter corresponding to the first letter of the word **product**.

Basic Exercise (continued)

(c) Write

$$1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

more compactly using summation notation and find [here](#) what familiar function's Taylor series it represents.

(d) Show that the formulas $\prod_{i=1}^n ca_i = c \prod_{i=1}^n a_i$ and $\prod_{i=1}^n (a_i \pm b_i) = \prod_{i=1}^n a_i \pm \prod_{i=1}^n b_i$ **do NOT hold** for products. What form does the first formula take instead in the case of products?

Double products function similarly to double sums.

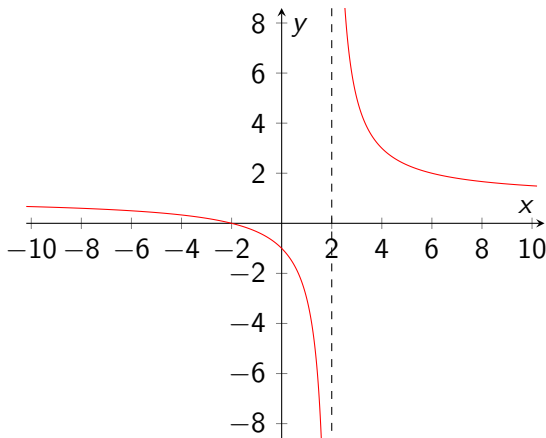
How to Solve Inequalities

General Solution Technique for Inequalities

Consider an inequality of the form $f(x) > 0$. (Note that all inequalities can be written in this or a similar form ($f(x) < 0$, $f(x) \geq 0$, $f(x) \leq 0$).)

To solve such an inequality, it is sufficient to determine when the values of the function f are positive. If f is continuous, it can only change sign at its zeros. If the function has discontinuities, the sign can also change at these points. These points can occur where the function is not defined or where the expression of a piecewise-defined function changes. There are no other types of points where the function's sign can change.

Thus, the zeros and discontinuities divide the domain of the function f into intervals where the function's sign does not change. For example, for the function shown in the figure, the sign can change only at $x = -2$ and $x = 2$, so the function has the same sign on the intervals $(-\infty, -2)$, $(-2, 2)$, and $(2, \infty)$.



The sign of the function on each interval can be determined by calculating the function value at a point within each interval or by inspecting the graph if one is available. We have thus outlined how to determine the intervals where the function value is positive or negative and, therefore, how to solve inequalities. This reasoning can be summarized in a clear algorithm:

- 1 Write the given inequality in the form $f(x) > 0$, $f(x) < 0$, $f(x) \geq 0$, or $f(x) \leq 0$.
- 2 Find the zeros and discontinuities of the function f .
- 3 Divide the domain of the function f into intervals based on the zeros and discontinuities.
- 4 Determine the sign of the function f on each of these intervals by calculating the function value at a point within each interval.
- 5 Formulate the solution.

Example

Solve the inequality

$$\frac{1}{x+2} \geq -\frac{1}{x-4}. \quad (3)$$

Solution

Since the inequality can also be equivalently written as

$$\frac{1}{x+2} + \frac{1}{x-4} \geq 0,$$

it is sufficient to determine when the function $f(x) = \frac{1}{x+2} + \frac{1}{x-4}$ is positive.

Solution (continued)

Discontinuities: $x = -2$ and $x = 4$. Zeros:

$$\frac{1}{x+2} + \frac{1}{x-4} = 0 \quad | \cdot (x+2)$$

$$1 + \frac{x+2}{x-4} = 0 \quad | \cdot (x-4)$$

$$x - 4 + x + 2 = 0$$

$$2x - 2 = 0$$

$$x = 1,$$

which is indeed a zero because it satisfies the condition for the equation's definition $x \neq -2$ and $x \neq 4$.

The sign of the function f thus remains the same on the intervals $(-\infty, -2)$, $(-2, 1)$, $(1, 4)$, and $(4, \infty)$.

Solution (continued)

Next we determine the sign of the function f on each of these intervals:

$(-\infty, -2)$: $f(-3) = -8/7 < 0$, so the function's sign is negative on this interval.

$(-2, 1)$: $f(0) = 1/4 > 0$, so the function's sign is positive on this interval.

$(1, 4)$: $f(2) = -1/4 < 0$, so the function's sign is negative on this interval.

$(4, \infty)$: $f(5) = 8/7 > 0$, so the function's sign is positive on this interval.

The answer includes the zero $x = 1$ because of the inequality. However, the discontinuities $x \neq -2$ and $x \neq 4$ are not included since the function is not defined at these values.

Therefore, the solution is $-2 < x \leq 1$ or $x > 4$.

Example

Solve the inequality $2^x < 8$.

It is sufficient to determine when $f(x) = 2^x - 8$ is negative.

Zeros:

$$2^x - 8 = 0$$

$$2^x = 8$$

$$2^x = 2^3$$

$$x = 3.$$

Discontinuities: none.

The function f thus has the same sign for $x < 3$ and $x > 3$.

$$x < 3: f(0) = 2^0 - 8 = 1 - 8 = -7 < 0.$$

$$x > 3: f(4) = 2^4 - 8 = 16 - 8 = 8 > 0.$$

Answer: $x < 3$.

Basic Exercise

Solve the inequality $x^2 - 4 > 0$ using the technique introduced above.

The technique for solving inequalities introduced in the previous example can be succinctly presented using a sign chart: construct a sign chart for the function f and deduce the solution from it. In this case, the sign chart would be as follows:

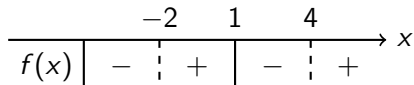


Figure: Sign chart for the function in Example (3).

When constructing a sign chart, determining the sign of the function over different intervals can be inferred from the graph in some cases. This is especially true for first and second-degree polynomials. For most functions, however, this approach can be cumbersome, so it is quicker to simply calculate the function value at each interval. In this case, it is best to choose values for which the calculations are straightforward.

Note that a sign chart is not necessary for solving the problem, but it can clarify the solution. Also, in the sign chart, it is helpful to distinguish between zeros and discontinuities to simplify the consideration of intervals.

Exercise

Solve the inequality $f(x) \geq 0$ based on the following information:

- (a) The function f has zeros at $x = -3$ and $x = \frac{2}{5}$.
- (b) The function f is continuous everywhere except at $x = -1$ and $x = 2$.
- (c) It is known that $f(-10) > 0$, $f(-2) < 0$, $f(0) < 0$, $f(1) > 0$, and $f(10) < 0$.

Basic Exercise

Solve the following inequalities:

(a) $x^2 - 3x - 10 \leq 0$

(b) $|2x + 5| > 2$

(c) $\frac{1}{x+2} \geq \frac{2}{x-3}$

(d) $\ln(x) < 0$.

For trigonometric equations, it is sufficient to solve the inequality over one period and then consider the multiples of this period. However, it is often easier to use the unit circle in solving, as will be done later. In such cases, the sign chart is replaced with the unit circle in the general technique for solving inequalities.

Example

Solve the inequality $\tan(x) > 1$.

Solution

Since the period of the tangent function is π , it is sufficient to determine when the function $f(x) = \tan(x) - 1$ is positive in the interval $[0, \pi)$ (or any other interval of length π).

Discontinuities: $x = \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$, with $x = \frac{\pi}{2}$ in the interval being considered.

Solution (continued)

Zeros:

$$\tan(x) - 1 = 0$$

$$\tan(x) = 1$$

$$(x = \arctan(1) + \pi n, n \in \mathbb{Z})$$

$$x = \frac{\pi}{4} + \pi n, n \in \mathbb{Z},$$

with $x = \pi/4$ in the interval being considered.

The function f thus has the same sign on the intervals $[0, \frac{\pi}{4})$, $(\frac{\pi}{4}, \frac{\pi}{2})$, and $(\frac{\pi}{2}, \pi)$.

$[0, \frac{\pi}{4})$: $f(0) = -1 < 0$, so f is negative on this interval.

$(\frac{\pi}{4}, \frac{\pi}{2})$: $f(\frac{\pi}{3}) = \sqrt{3} - 1 > 0$, so f is positive on this interval.

$(\frac{\pi}{2}, \pi)$: $f(\frac{2\pi}{3}) = -\sqrt{3} - 1 < 0$, so f is negative on this interval.

The inequality is thus satisfied in the interval $(\frac{\pi}{4}, \frac{\pi}{2})$ and its π -multiples, so the solution is $\frac{\pi}{4} + \pi n < x < \frac{\pi}{2} + \pi n, n \in \mathbb{Z}$.

Basic Exercise

Solve the inequality $\sin(x) > \sqrt{3}/2$.

Finally, it should be noted that compound inequalities, such as

$$2x - 1 < x < x^2,$$

are solved by separately solving the inequalities $2x - 1 < x$ and $x < x^2$.

Elementary Functions

Polynomials

Polynomials are functions that can be written in the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0 \quad (4)$$

or more compactly using summation notation as

$$f(x) = \sum_{i=0}^n a_i x^i,$$

where $a_i \in \mathbb{R}$ and $n \in \mathbb{Z}_{\geq 0}$. The number n is called the **degree** of the polynomial (assuming $a_n \neq 0$).

A polynomial equation of the form

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0 = 0$$

is called an **n -th degree equation**.

For example, $f(x) = x^2 + 1$ is a second-degree polynomial and $g(x) = 4$ is a zero-degree polynomial, since[†] $4 = 4x^0$.

[†]With polynomials, it is conventional to calculate the exponent before placing zero to avoid the indeterminate form 0^0 .

The term polynomial comes from the fact that expressions of the form ax^m , $a \in \mathbb{R}_{\neq 0}$, $m \in \mathbb{Z}_{>0}$ are called **monomials**; expressions of the form $ax^m + bx^n$, $a, b \in \mathbb{R}_{\neq 0}$, $m, n \in \mathbb{Z}_{>0}$ are called **binomials**, and more generally, a sum of an arbitrary number of such terms is called a polynomial[†].

For example, 3 , x^2 , and $3x^2$ are monomials, $3 + 2$, $x^2 + 2$, and $3x^2 - 4x$ are binomials, and all of these are polynomials.

[†]Poly is an ancient Greek word meaning many. The prefix poly- thus corresponds to the prefix multi-, just as mono- means single and bi- means two.

Example

Let $f(x) = 4x^3 + 6$ and $g(x) = 6x^4 - x^3 + x^2$. Determine the coefficients and degrees of the polynomials $f(x) + g(x)$ and $f(x)g(x)$.

Solution

$$\begin{aligned}f(x) + g(x) &= (4x^3 + 6) + (6x^4 - x^3 + x^2) \\&= 6x^4 + 3x^3 + x^2 + 6 \\&= 6x^4 + 3x^3 + x^2 + 0x + 6 \\&= \underbrace{6}_{a_4}x^4 + \underbrace{3}_{a_3}x^3 + \underbrace{1}_{a_2}x^2 + \underbrace{0}_{a_1}x + \underbrace{6}_{a_0}\end{aligned}$$

$$\begin{aligned}\text{and } f(x)g(x) &= (4x^3 + 6)(6x^4 - x^3 + x^2) \\&= 4x^3(6x^4 - x^3 + x^2) + 6(6x^4 - x^3 + x^2) \\&= 24x^7 - 4x^6 + 4x^5 + 36x^4 - 6x^3 + 6x^2 \\&= \underbrace{24}_{a_7}x^7 + \underbrace{(-4)}_{a_6}x^6 + \underbrace{4}_{a_5}x^5 + \underbrace{36}_{a_4}x^4 + \underbrace{(-6)}_{a_3}x^3 + \underbrace{6}_{a_2}x^2.\end{aligned}$$

The degree of the polynomial $f + g$ is 4 and the degree of the polynomial fg is 7. How are these degrees obtained from the degrees of the original polynomials?

As illustrated in the previous example, in general, the degree of the sum of polynomials f and g is the maximum of the degrees of f and g , and the degree of the product of polynomials f and g is the sum of the degrees of f and g .

Basic Exercise

If the degrees of polynomials f and g are n and m , what is the degree of their product fg and why?

A first-degree polynomial $f(x) = ax + b$, where $a \neq 0$, always has exactly one real root.

A second-degree polynomial $f(x) = ax^2 + bx + c$, where $a \neq 0$, can have at most two real roots given by the formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

where the number of real roots is determined by the value of the discriminant $D = b^2 - 4ac$, that is,

$$\begin{cases} \text{there are two (distinct) real roots if } D > 0, \\ \text{there is a single real root if } D = 0, \\ \text{there are no real roots if } D < 0. \end{cases}$$

In the case of a single root, the root $x = x_1$ is considered a “double root” because $(x - x_1)^2 = 0$.

This formulation generalizes to all polynomials except for the constant polynomial $p(x) = 0$.

A polynomial of degree n (non-zero) has at most n real roots.

Remark. In fact, a more significant result known as the **Fundamental Theorem of Algebra** (or more precisely, its consequence) states that

A non-constant polynomial of degree n has exactly n complex roots (counting their multiplicities).

The roots that are not real are complex, i.e., of the form $a + bi$, where $a, b \in \mathbb{R}$ and i satisfies $i^2 = -1$. Real numbers are complex numbers where $b = 0$.

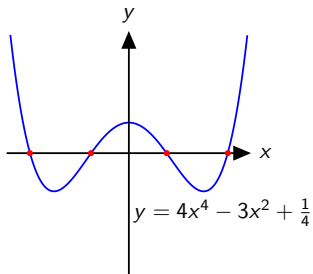
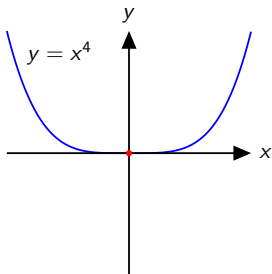
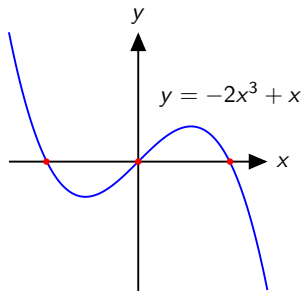
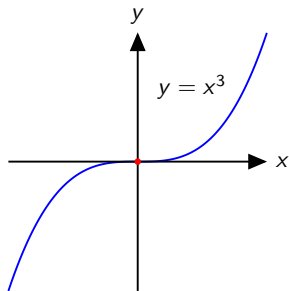
Basic Exercise

Find second-degree polynomials that have (a) two (b) one (c) no roots, and plot their graphs. How does the number of roots appear in the graph?

The roots of second-degree polynomials are always found using the quadratic formula. There are also formulas for the roots of third and fourth-degree polynomials, but they are so complex that they are not used for manual calculations.

It has been proven that there are no general formulas for the roots of polynomials of degree five or higher. However, higher-degree ($n \geq 3$) polynomials can still be solved manually in certain situations. For example, in the case of $5x^8 = 10$, the solution is $x = \pm\sqrt[8]{2}$. Next, we will review three such techniques, each of which aims to reduce the original problem to solving first and second-degree polynomials.

Roots of polynomial functions



Technique 1

The first technique is based on the **zero-product property**, which states

$$ab = 0 \iff a = 0 \text{ or } b = 0.$$

The direction $\implies$ is the essential one here and means that the product of real numbers can be zero only if one (or both) of the factors is zero. The direction $\impliedby$ is self-evident. It is important to note that this property does not apply to any other number except zero.

Basic Exercise

Find 7 different cases where $ab = 1$.

The zero-product property also applies to multiple numbers and to more complex expressions. For example, it holds that

$$abc = 0 \iff a = 0 \text{ or } b = 0 \text{ or } c = 0$$

and

$$f(x)g(x) = 0 \iff f(x) = 0 \text{ or } g(x) = 0.$$

Now let's look at an example of applying the zero-product property to solve a polynomial equation.

Example

Solve the equation $x^5 - 3x^4 + 2x^3 = 0$.

Solution

By factoring out the common factor and applying the zero-product property, we get

$$\begin{aligned}x^5 - 3x^4 + 2x^3 &= 0 \\x^3(x^2 - 3x + 2) &= 0 \\x^3 &= 0 \quad \text{or} \quad x^2 - 3x + 2 = 0 \\x &= 0 \quad \text{or} \quad x = 1 \quad \text{or} \quad x = 2.\end{aligned}$$

Answer: $x = 0$, $x = 1$, or $x = 2$.

Why didn't the previous technique work in the case of $x^5 - 3x^4 + x^3 = 2$?

Basic Exercise

Solve the equation $x^2 + 2x = 4$.

A more complex expression than x can be factored out, but these are harder to notice.

Example

By factoring out the common factor and applying the zero-product property, we get

$$\begin{aligned}x^3 - 2x^2 + x - 2 &= 0 \\x^2(x - 2) + (x - 2) &= 0 \\(x^2 + 1)(x - 2) &= 0 \\x^2 + 1 = 0 \quad \text{or} \quad x - 2 &= 0 \\ \text{no solution} \qquad \qquad \qquad x &= 2.\end{aligned}$$

Answer: $x = 2$.

Note that this solution technique is not limited to polynomials but also works, for example, in the case of $e^x \sin(x) = 0$.

Technique 2

The second technique is based on the fact that all polynomials can be alternatively written as products of monomials based on its roots.

Example

Consider the solutions of a quadratic equation $x = 1$ or $x = -1$. Therefore, we also have $x - 1 = 0$ or $x + 1 = 0$. Hence, based on the zero-product property, we also have $(x - 1)(x + 1) = 0$, which can be further written as $x^2 - 1 = 0$, whose solutions are clearly $x = \pm 1$.

The key point in this reasoning is that the polynomial $x^2 - 1$ can be expressed as the product $(x - 1)(x - (-1))$ based on its roots.

This is not an exceptional case; it generally holds true as demonstrated on the following slide.

If a polynomial f of degree n has roots $x_1, x_2, \dots, x_n$, it can be written in the form

$$f(x) = a(x - x_1)(x - x_2) \cdots (x - x_n)$$

or

$$f(x) = a \prod_{i=1}^n (x - x_i),$$

where a is the coefficient of the highest degree term in the polynomial f . (Some of the roots may be complex.)

The coefficient a needs to be included if the coefficient of the highest degree term is not 1.

Basic Exercise

- (a) Write the polynomial $f(x) = x^2 - 4x - 5$ in terms of its roots.
- (b) Write the polynomial $f(x) = 3x^2 - 12x - 15$ in terms of its roots.
- (c) Find a polynomial whose roots are $x = 7$ and $x = -3/2$, and write it in the form $ax^2 + bx + c = 0$, where a , b , and c are integers.

The previous result can also be expressed in a more convenient form for our purposes.

If the polynomial f of degree n has roots $x_1, x_2, \dots, x_m$, it can be written in the form

$$f(x) = g(x)(x - x_1)(x - x_2) \cdots (x - x_m) \quad (5)$$

or

$$f(x) = g(x) \prod_{i=1}^m (x - x_i),$$

where $g(x)$ is a polynomial of degree $n - m$.

This result can be used to reduce solving a higher degree equation to solving a lower degree equation. For example, if we know that the polynomial $f(x) = x^3 + x^2 + x + 1$ has a root $x = -1$, it can be written as

$$x^3 + x^2 + x + 1 = (x + 1)g(x),$$

so the remaining roots are obtained as the roots of the polynomial g based on the zero-product property. However, this leaves two questions open: How is the polynomial g determined, and how do we find the required root $x = -1$?

If you are familiar with **polynomial division**, g can be found by dividing

$$g(x) = \frac{x^3 + x^2 + x + 1}{x + 1}.$$

Alternatively, g can be determined using the technique shown in the following example, which is worth learning even if you are familiar with polynomial division, as similar reasoning can be used in other cases.

Roots can be found by trial. In the problems of this course, the situations are simple enough that the required root is usually a simple integer, but in more complex cases, the possible rational roots (and thus integer roots) of a polynomial with integer coefficients can be restricted to a relatively small set using the **Rational Root Theorem**, making the trial process more manageable. Alternatively, if you can graph the function, you can get rough estimates of the roots from the graph, which can then be verified by substitution.

Example

Find the roots of the polynomial $f(x) = -3 + 4x + 2x^2 - 4x^3 + x^4$ given that $f(-1) = 0$.

Solution

$f(-1) = 0$, so $x = -1$ is a root. It also holds that

$$-3 + 4 \cdot 1 + 2 \cdot 1^2 - 4 \cdot 1^3 + 1^4 = 0,$$

so $x = 1$ is also a root. Thus,

$$\begin{aligned} f(x) &= g(x)(x-1)(x+1) \\ &= (b_0 + b_1x + b_2x^2)(x-1)(x+1) \\ &= (b_0 + b_1x + b_2x^2)(x^2 - 1) \\ &= b_0x^2 + b_1x^3 + b_2x^4 - b_0 - b_1x - b_2x^2 \\ &= -b_0 - b_1x + (b_0 - b_2)x^2 + b_1x^3 + b_2x^4 \end{aligned}$$

$$\begin{aligned} \text{or } -3 + 4x + 2x^2 - 4x^3 + x^4 \\ = -b_0 - b_1x + (b_0 - b_2)x^2 + b_1x^3 + b_2x^4. \end{aligned}$$

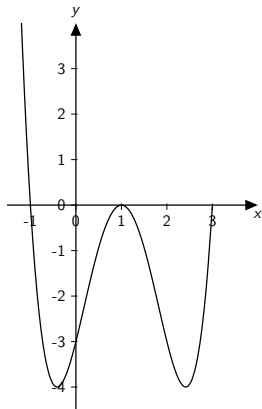


Figure:

$$\begin{aligned} f(x) \\ = -3 + 4x + 2x^2 - 4x^3 + x^4. \end{aligned}$$

Solution (continued)

The coefficients must be the same on both sides of the equation, so

$$\begin{cases} -b_0 = -3 \\ -b_1 = 4 \\ b_0 - b_2 = 2 \\ b_1 = -4 \\ b_2 = 1 \end{cases} \iff \begin{cases} b_0 = 3 \\ b_1 = -4 \\ b_2 = 1 \end{cases},$$

so

$$g(x) = 3 - 4x + x^2.$$

The remaining roots are:

$$\begin{aligned} 3 - 4x + x^2 &= 0 \\ x &= 1 \quad \text{or} \quad x = 3. \end{aligned}$$

Answer: $x = -1$, $x = 1$, or $x = 3$.

In the previous solution, the function g could also be determined by performing polynomial division:

$$g(x) = \frac{f(x)}{(x-1)(x+1)} = \frac{x^4 - 4x^3 + 2x^2 + 4x - 3}{x^2 - 1}.$$

Note also that although the 5×3 system of equations can be solved based on three equations, it is still necessary to check that the remaining two equations are also satisfied. If they are not, the system would not have a solution. This will be discussed further towards the end of the course when solving systems of equations more generally.

Basic Exercise

Solve the equation

$$-\lambda^3 + 3\lambda^2 + \lambda - 3 = 0.$$

Start by verifying that $\lambda = 3$ is a root. How does this help in finding the general solution?

Technique 3

Let's review the third and final technique. It relies on the fact that certain types of equations can be simplified to quadratic equations through an appropriate substitution.

Example

Solve the equation $x^4 + x^2 - 6 = 0$.

Solution

$$x^4 + x^2 - 6 = 0$$

$$(x^2)^2 + x^2 - 6 = 0 \quad | \text{ Let } x^2 = y.$$

$$y^2 + y - 6 = 0$$

$$y = -3 \quad \text{or} \quad y = 2 \quad | \quad y = x^2$$

$$x^2 = -3 \quad \text{or} \quad x^2 = 2$$

$$\text{no solution} \quad \text{or} \quad x = \pm\sqrt{2}$$

Answer: $x = \pm\sqrt{2}$.

The previous solution can also be handled without substitution. In that case, the solutions to the quadratic equations are directly $x^2 = -3$ or $x^2 = 2$. However, substitution can make the solution clearer.

Note that the previous technique is not limited to polynomials; it can also be used, for example, for the equation $\sin^2(x) + \sin(x) = 1$ and more generally for any equation of the form

$$a(f(x))^{2n} + b(f(x))^n + c = 0.$$

(What would be the substitution that transforms this equation into a quadratic equation?)

Basic Exercise

Solve the following equations:

(a) $x^6 + 7x^3 - 8 = 0$

(b) $x^3 + x^2 - 5x = 0$

(c) $x^3 + x^2 - 5x = 5$

(d) $(x - 3)(x + 7)(x - 4) = 0.$

Rational functions

Review of Rules for Expressions Involving Division

The rules

$$(1) \frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$$

$$(2) \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

$$(3) \frac{a}{b} : \frac{c}{d} = \frac{ad}{bc}$$

are introduced for handling fractions where a , b , c , and d are integers. However, they also apply directly to division involving real numbers. Thus, they also apply when variables and functions are involved, as long as division by zero is avoided.

The first rule might initially seem confusing, but it simply encapsulates the fact that for addition or subtraction, the numbers must first be converted to a common denominator. Since the division rule is clearly less intuitive than the others, let's review it a bit further.

Example

$$\frac{\left(\frac{1}{3}\right)}{4} = \frac{\left(\frac{1}{3}\right)}{\left(\frac{4}{1}\right)} = \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{3 \cdot 4} = \frac{1}{12}.$$

The division formula is easily remembered by the following formulation: **dividing is the same as multiplying by the reciprocal**. This means, written as an equation:

$$\frac{a}{b} = a \cdot \frac{1}{b}.$$

Based on this and the common denominator, it can further be explained why it holds that

$$-\frac{a}{b} = \frac{-a}{b} = \frac{a}{-b}.$$

Also recall that the reciprocal of a number is the number such that its product with the original number equals 1. Thus, the reciprocal of $a \neq 0$ is $1/a$ and the reciprocal of a fraction a/b , with $a, b \neq 0$, is b/a .

Basic Exercise

Simplify the following expressions:

$$(a) \ 2 : \frac{1}{4}$$

$$(b) \ \frac{\left(\frac{3}{4}\right)}{\left(\frac{6}{5}\right)}$$

$$(c) \ \frac{\left(\frac{x^2}{6}\right)}{\left(\frac{3}{x}\right)}, \ x \neq 0$$

$$(d) \ \frac{3}{e^{2x}} : \frac{2}{e^{-3x}}.$$

Also, recall that for division the formula

$$\frac{a+b}{c} = \frac{a}{c} + \frac{b}{c},$$

follows from the distributive property $(a+b)c = ac + bc$.

Basic Exercise

Simplify the expression for the function

$$f: \mathbb{R}_{\neq 0} \rightarrow \mathbb{R}, \quad f(x) = \frac{x^4 + 2x^3 + 3x^2 + 4x + 5}{x^2}.$$

Let's also take a small test on simplification. If the following problem raises any doubts, it is advisable to review Appendix 6.5, which explains the logic of simplification in more detail.

Basic Exercise

Which of the following expressions can be simplified by dividing by 2 (so that it does not remain in the denominator):

(a) $\frac{2+x}{2}$

(b) $\frac{1+2x}{2}$

(c) $\frac{2+2x}{2}$

(d) $\frac{2}{2+x}$

(e) $\frac{2}{2+2x}$

(f) $\frac{2+2x}{2+x}$

(g) $\frac{2+x}{1+2x}$

(h) $\frac{1+2x}{1+2x}$

(i) $\frac{2+2x}{2+2x}$

(j) $\frac{2(1+x)}{2(1+x)}$

(k) $\frac{\cos(2x)}{2}$

(l) $\frac{2\cos(x)}{2}$

(m) $\frac{\cos^2(x)}{2}$

(n) $\frac{2^x}{2}$

(o) $\frac{(\log_2(2x^2))^2}{2}?$

And then to the main point...

Functions of the form $f(x) = P(x)/Q(x)$, where P and Q are polynomials, are called **rational functions**. They are defined everywhere except at the zeros of the denominator, i.e., where $Q(x) = 0$.

Rational functions are divisions of polynomials. Thus, the arithmetic rules reviewed earlier are used in their manipulation. However, the previous chapter did not discuss rational expressions, as the arithmetic rules covered apply to other types of expressions as well, such as $\sin(x)/x$, which is not a rational expression.

Example

Determine the domain restrictions and simplify the following rational expressions:

$$(a) \frac{x+3}{x^2-1} - \frac{x+1}{x^2-x}$$

$$(b) \frac{ab-b^2}{4a^2-b^2} \cdot \frac{2a^2+ab}{b-a}.$$

Solution

In each case, the domain restriction is only limited by the requirement that division by zero is not allowed.

(a) Domain restriction:

$$\begin{aligned}x^2 - 1 \neq 0 \text{ and } x^2 - x \neq 0 &\iff x \neq \pm 1 \text{ and } (x \neq 0 \text{ and } x \neq 1) \\&\iff x \neq \pm 1 \text{ and } x \neq 0.\end{aligned}$$

Simplification:

$$\begin{aligned}\frac{x+3}{x^2-1} - \frac{x+1}{x^2-x} &= \frac{(x^2-x)(x+3) - (x+1)(x^2-1)}{(x^2-1)(x^2-x)} \\&= \frac{x^3 + 3x^2 - x^2 - 3x - (x^3 - x + x^2 - 1)}{(x^2-1)(x^2-x)} \\&= \frac{x^2 - 2x + 1}{(x^2-1)(x^2-x)} = \frac{(x-1)^2}{x(x^2-1)(x-1)} \\&= \frac{x-1}{x(x^2-1)} = \frac{x-1}{x(x+1)(x-1)} \\&= \frac{1}{x(x+1)}, \quad x \notin \{0, \pm 1\}.\end{aligned}$$

Solution

- (b) Domain restriction: $4a^2 - b^2 \neq 0$ and $b - a \neq 0$, i.e., $b \neq \pm 2a$ and $b \neq a$. Simplification:

$$\begin{aligned}\frac{ab - b^2}{4a^2 - b^2} \cdot \frac{2a^2 + ab}{b - a} &= \frac{b(a - b)}{4a^2 - b^2} \cdot \frac{a(2a + b)}{b - a} \\&= \frac{b(a - b)}{(2a + b)(2a - b)} \cdot \frac{a(2a + b)}{-(a - b)} \\&= -\frac{ab}{2a - b} \\&= \frac{ab}{b - 2a}.\end{aligned}$$

Basic Exercise

Determine the domain restrictions and simplify the following rational expressions:

$$(a) \frac{x^2 - 1}{x + 1}$$

$$(b) \frac{x^3 - 6x^2 + 9x}{5x^2 - 15x}$$

$$(c) \frac{xy^2 - x}{y^2} : \frac{xy + x}{y}.$$

Since **division can be zero only when its numerator is zero**, the zeros of a rational function $f(x) = P(x)/Q(x)$ are found at the zeros of its numerator $P(x)$. However, they must also belong to the domain of the function, i.e., satisfy the condition $Q(x) \neq 0$. Thus, to solve the rational equation $P(x)/Q(x) = 0$, it is sufficient to find the zeros of the polynomial P and check that they satisfy the domain condition of the equation. If the rational equation is not already in the form $P(x)/Q(x) = 0$, such as the equation

$$\frac{-2}{x^2 - 1} = \frac{1}{x + 1},$$

it can be transformed into this form by suitable expansion.

A rational equation can also be solved by first eliminating the denominators. This is probably the more illustrative method. Even then, it is important to verify that the proposed solutions satisfy the domain condition. The following example demonstrates both methods.

Example

Solve the equation

$$\frac{-2}{x^2 - 1} = \frac{1}{x + 1}.$$

Solution

Method 1: Solve the equation as described above by writing it in the form $P(x)/Q(x) = 0$ and finding the zeros of the function P .

Domain restriction:

$$x^2 - 1 \neq 0 \text{ and } x \neq -1 \iff x \neq \pm 1 \text{ and } x \neq -1 \iff x \neq \pm 1.$$

Solution (continued)

Solution(s):

$$\begin{aligned}\frac{-2}{x^2 - 1} &= \frac{1}{x + 1} \\ \frac{-2}{(x + 1)(x - 1)} - \frac{1}{x + 1} &= 0 \\ \frac{-2}{(x + 1)(x - 1)} - \frac{x - 1}{(x + 1)(x - 1)} &= 0 \\ \frac{-2 - (x - 1)}{(x + 1)(x - 1)} &= 0 \\ \frac{-x - 1}{x^2 - 1} &= 0 \\ -x - 1 &= 0 \\ x &= -1.\end{aligned}$$

Since $x = -1$ does not satisfy the domain restriction of the equation, it is not a solution.

Answer: no solution.

Solution (continued)

Method 2: In this solution, the denominators are first eliminated from the equation.

Domain restriction:

$$x^2 - 1 \neq 0 \text{ and } x \neq -1 \iff x \neq \pm 1 \text{ and } x \neq -1 \iff x \neq \pm 1.$$

Solution(s):

$$\begin{aligned}\frac{-2}{x^2 - 1} &= \frac{1}{x + 1} && | \cdot (x^2 - 1) \\ -2 &= \frac{x^2 - 1}{x + 1} \\ -2 &= \frac{(x + 1)(x - 1)}{x + 1} \\ -2 &= (x - 1) \\ x &= -1.\end{aligned}$$

Since $x = -1$ does not satisfy the domain restriction of the equation, it is not a solution.

Answer: no solution.

Basic Exercise

Solve the following equations:

$$(a) \frac{1}{x} = 0$$

$$(b) \frac{x^2 + 3x - 4}{x - 1} = 0$$

$$(c) \frac{3}{1 - x} = \frac{x + 1}{x - 3}.$$

Trigonometric Functions

Instead of degrees, the unit of angle is often measured in radians. The relationship between radians and degrees can be expressed as

$$360^\circ = 2\pi \text{ rad},$$

which practically means that a full circle corresponds to 2π , or approximately 6.28 radians. Often the previous equation is given in the equivalent form $180^\circ = \pi$, but it may be easier to remember the situation corresponding to a full circle compared to a half circle. Note particularly that in a full circle, the number of radians corresponds to the circumference of a circle with radius 1.

An angle is a dimensionless quantity, so its unit does not need to be written out explicitly, and generally, this is how radians are handled. In the future, radians will only be written out in exceptional cases. However, if both degrees and radians are considered simultaneously, it is advisable to indicate the unit of degrees to avoid confusion.

From the previous equation, we can infer how to convert any angle from degrees to radians and vice versa. For example, by dividing the equation by two, we get the equation $180^\circ = \pi$. Other simple angles such as $\frac{\pi}{2}$, $\frac{\pi}{3}$, $\frac{\pi}{4}$, and $\frac{\pi}{6}$ can be quickly deduced in a similar manner. For instance, by dividing the previous equation by three, we obtain $\frac{\pi}{3} = 60^\circ$.

In less intuitive cases, it might be clearer to first derive the formulas

$$1^\circ = \frac{\pi}{180} \text{ rad} \quad \text{and} \quad 1 \text{ rad} = \frac{180^\circ}{\pi}$$

and then use them, as illustrated in the following example.

Example

Convert the following angles from degrees to radians or from radians to degrees:

(a) π°

(b) 540°

(c) $\frac{\pi}{5}$ rad

(d) 6 rad.

Solution

(a)

$$\pi^\circ = \pi \cdot 1^\circ = \pi \cdot \frac{\pi}{180} = \frac{\pi^2}{180} \approx 0.055$$

(b) This can be quickly inferred. It is one and a half full circles, so

$$540^\circ = \frac{3}{2} \cdot 360^\circ = \frac{3}{2} \cdot 2\pi = 3\pi.$$

(c) This can be quickly inferred. $\frac{\pi}{5}$ is one-fifth of the angle corresponding to a half circle, so

$$\frac{\pi}{5} = \frac{1}{5} \cdot \pi = \frac{1}{5} \cdot 180^\circ = 36^\circ.$$

Solution (continued)

(d) Thus, this angle corresponds to slightly less than a full circle.

$$6 \text{ rad} = 6 \cdot (1 \text{ rad}) = 6 \cdot \frac{180^\circ}{\pi} = \frac{1080^\circ}{\pi} \approx 344^\circ,$$

which seems reasonable in magnitude.

Basic Exercise

Convert the following angles from degrees to radians or from radians to degrees:

(a) 10°

(b) $\sqrt{10}^\circ$

(c) $\frac{\pi}{8}^\circ$

(d) 720°

(e) 10 rad

(f) $\sqrt{10} \text{ rad}$

(g) $\frac{\pi}{8} \text{ rad}$

(h) 720 rad .

Calculate both exact values and approximate values.

Why are radians used? The reason is that radians are the unit of angle in which formulas are most simplified. In this sense, it is the angle unit that is usually preferred in mathematics. For example, the formula for the length b of an arc of a circle with radius r and the derivative formula of sine are in radians given by

$$b = xr \quad \text{and} \quad D \sin(x) = \cos(x),$$

whereas in degrees, the formulas are given by

$$b = \frac{\pi}{180^\circ} xr \quad \text{and} \quad D \sin(x) = \frac{\pi}{180^\circ} \cos(x).$$

In the case of degrees, the formulas include an unpleasant correction factor $\frac{\pi}{180^\circ}$. This, in turn, complicates all subsequent formulas that require the previous ones. For this reason, the derivative formulas of trigonometric functions are practically always given only in radians. This topic is discussed in more detail in Appendix 6.7.

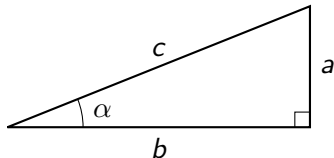
When should radians be used? In some situations, such as discussing the steepness of a hill, degrees may be used for clarity, but generally, radians should be used because there is a risk that the formula might not apply when using degrees. There may be formulas where degrees must be used, but if in doubt, always use radians. In this and subsequent course materials, radians are almost always used, and degrees are used only for illustrative purposes in some cases.

Trigonometry of Right Triangles

For a right triangle (see figure below), the trigonometric identities are

$$\sin(\alpha) = \frac{a}{c}, \quad \cos(\alpha) = \frac{b}{c} \quad \text{and} \quad \tan(\alpha) = \frac{a}{b}. \quad (6)$$

A trigonometric equation always has infinitely many solutions. If this is not familiar, you can illustrate the concept by calculating values like



$\dots, \cos(-420^\circ), \cos(-60^\circ), \cos(60^\circ), \cos(420^\circ), \dots, \cos(60^\circ + 360^\circ n), \dots$

where $n \in \mathbb{Z}$. Each of these expressions evaluates to $\frac{1}{2}$, meaning that all these angles are solutions to the equation $\cos(x) = \frac{1}{2}$. When solving a trigonometric equation, one needs to find all such angles. However, if we are solving for an angle in a right triangle, the situation simplifies significantly. The non-right angles of a right triangle are always in the interval $(0^\circ, 90^\circ)$, and it can be shown that the equations (6) have exactly one solution in this interval. This is precisely the solution obtained with a calculator.

In addition to sine, cosine, and tangent, there are three other trigonometric functions: secant (sec), cosecant (csc), and cotangent (cot). For these functions in a right triangle, the identities are

$$\sec(\alpha) = \frac{c}{b}, \quad \csc(\alpha) = \frac{c}{a} \quad \text{and} \quad \cot(\alpha) = \frac{b}{a}.$$

These functions are rarely encountered because they are not generally needed. For example, if the sides a and b of a right triangle are known and one wants to calculate the angle α , it can be done using both tangent and cotangent, but tangent is usually preferred.

Why are there exactly six trigonometric functions? There is a certain dependency between the angles of a right triangle and the **ratios of the sides** of the triangle. Trigonometric functions describe these dependencies. Since there are six different **ratios** that can be formed from the three sides, there are six trigonometric functions.

Law of Sines and Law of Cosines

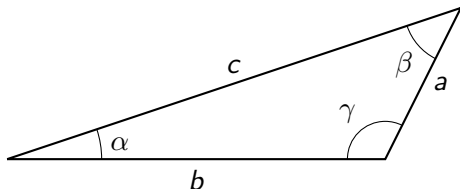
For an arbitrary triangle (see figure below), the trigonometric identities are

$$\frac{a}{\sin(\alpha)} = \frac{b}{\sin(\beta)} = \frac{c}{\sin(\gamma)}$$

and

$$a^2 = b^2 + c^2 - 2bc \cos(\alpha),$$

and these facts are known as the **Law of Sines** and the **Law of Cosines**. These can be used to handle triangles for which the formulas from the previous section do not apply.



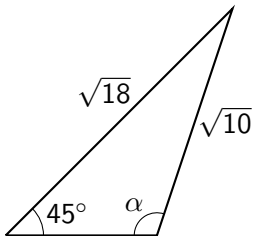
In an arbitrary triangle, the angle to be solved is always in the interval $(0^\circ, 180^\circ)$. The cosine rule has exactly one solution in this interval, which is directly obtained with a calculator, just as in the case of a right triangle.

However, the situation becomes more complex with the Law of Sines, because the sine rule typically has two solutions in the interval $(0^\circ, 180^\circ)$, one in the interval $(0^\circ, 90^\circ)$ and the other in the interval $(90^\circ, 180^\circ)$. (In the case of 90° , there is no second solution.) The calculator will provide the solution in the interval $(0^\circ, 90^\circ)$, so if the angle to be solved is greater than 90° , the angle given by the calculator is not correct. The correct angle is obtained by the expression

$180^\circ - \text{the angle given by the calculator.}$

Example

Determine the size of angle α in the triangle shown.



Solution

Since two sides and one angle are known, the Law of Sines should be used. According to this law, we have...

Solution (continued)

$$\frac{\sqrt{10}}{\sin(45^\circ)} = \frac{\sqrt{18}}{\sin(\alpha)}$$

$$\sin(\alpha) = \frac{\sqrt{18}}{\sqrt{10}} \sin(45^\circ) \quad \left| \quad \sin(45^\circ) = \frac{1}{\sqrt{2}} \right.$$

$$\sin(\alpha) = \sqrt{\frac{18}{10}} \frac{1}{\sqrt{2}}$$

$$\sin(\alpha) = \frac{3}{\sqrt{10}}.$$

The calculator provides the solution to this equation as

$$\alpha = \sin^{-1} \left(\frac{3}{\sqrt{10}} \right) \approx 71.6^\circ.$$

This clearly cannot be the correct solution (Why not?), so the correct solution is

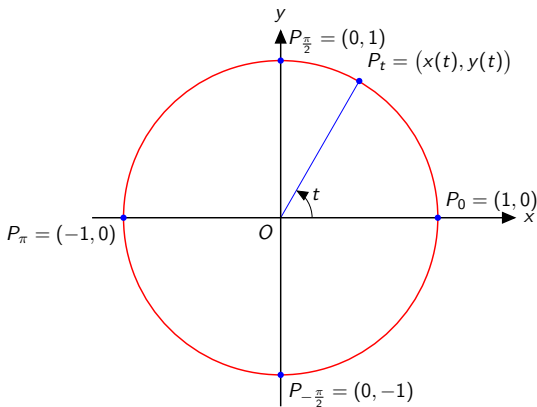
$$\alpha = 180^\circ - \sin^{-1} \left(\frac{3}{\sqrt{10}} \right) \approx 108.4^\circ.$$

Unit Circle

The **unit circle** refers to a circle with radius 1 centered at the origin. It allows us to define sine and cosine for all possible angles. An angle is understood as a directional angle (denoted t in the figure below) that starts from the positive x -axis and has a positive direction counterclockwise. An angle can also be measured clockwise, in which case its value is negative. Furthermore, an angle can also exceed a full circle, resulting in an angle whose absolute value is greater than 2π . For example, an angle of -4π corresponds to two full clockwise rotations. Thus, an angle corresponding to a full rotation can be chosen in infinitely many ways. Usually, when considering one full rotation, we look at the interval $(-\pi, \pi]$ or $[0, 2\pi)$. One endpoint is left open because otherwise, the angle would be considered with two different values.

The equation of the unit circle is $x^2 + y^2 = 1$. Points that satisfy this equation, i.e., points on the circumference of the circle, are called **circumference points** of the unit circle. In the figure below, the circumference point corresponding to angle t is denoted by P_t . For example, the point $(0, 1)$ corresponds to the angle $\frac{\pi}{2}$. The same circumference point corresponds to infinitely many angles. For example, $P_\pi = P_{3\pi} = P_{-3\pi} = (-1, 0)$.

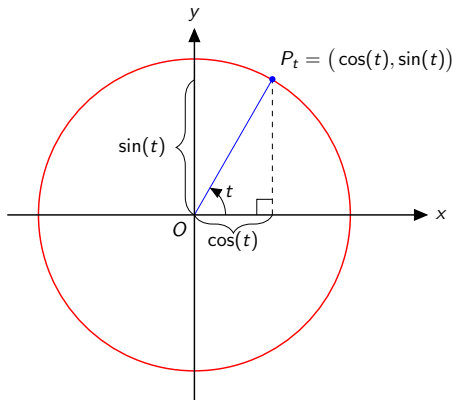
The accompanying figure shows the unit circle, angle t , and the circumference points corresponding to the angles 0 , t , $\frac{\pi}{2}$, π , and $-\frac{\pi}{2}$. The coordinates of the circumference point P_t corresponding to angle t depend on the value of t . If t is considered as time, then P_t indicates the location of the circumference point at each moment in time.



The notation t is used for the angle rather than x , because x is reserved for the values of the horizontal axis.

Definitions of Trigonometric Functions

The functions $\sin$ and $\cos$ are defined using the unit circle as shown in the figure below.



The cosine of angle t , $\cos(t)$, is the x -coordinate of the point on the unit circle corresponding to the angle t , and the sine of angle t , $\sin(t)$, is the y -coordinate of the same point.

Thus, the point on the unit circle corresponding to angle t is $(\cos(t), \sin(t))$. For example, the value of the cosine at $t = 0$ is $\cos(0) = 1$ and the value of the sine is $\sin(0) = 0$.

Using the functions $\sin(t)$ and $\cos(t)$, we can also define other trigonometric functions, the most important of which is the tangent. Its definition is

$$\tan(t) = \frac{\sin(t)}{\cos(t)}. \quad (7)$$

The definitions of the other three trigonometric functions are

$$\sec(t) = \frac{1}{\cos(t)}, \quad \csc(t) = \frac{1}{\sin(t)} \quad \text{and} \quad \cot(t) = \frac{\cos(t)}{\sin(t)}.$$

Note that these are simply the reciprocals of cosine, sine, and tangent for each value of t (not the inverse functions). Therefore, these functions are not strictly necessary, and hence $\sec$, $\csc$, and $\cot$ are generally not discussed further.

Understanding the definitions of sine and cosine is extremely important in trigonometry because a large part of the understanding relies on them.

You must be able to show roughly using the unit circle

- 1 the values of sine and cosine for any angle
- 2 the angles corresponding to any values of sine and cosine.

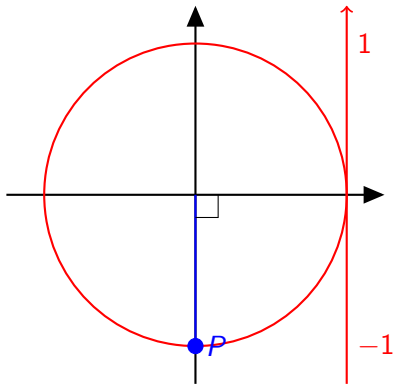
Determining exact values is only possible in special cases, but you should be able to show a rough estimate from the figure in any case. You should also understand the interpretation of tangent using the unit circle. Let us now look at some important properties of trigonometric functions.

Example

- (a) Determine the exact values of the sine, cosine, and tangent of the angle $-\frac{\pi}{2}$ based on the unit circle.
- (b) Determine an estimate for the sine, cosine, and tangent of the angle -50° based on the unit circle.

Solution

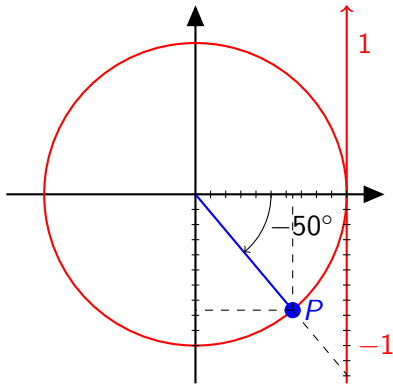
(a) Start by drawing a diagram.



The sine and cosine of an angle are the x and y coordinates of the corresponding point P on the unit circle. Thus, we have $\cos(-\frac{\pi}{2}) = 0$ and $\sin(-\frac{\pi}{2}) = -1$. The value of the tangent would be the y -coordinate of the intersection point of the red line and the extension of the blue line. Since they do not intersect, it does not exist. How can you deduce this also based on the equation $\tan(x) = \sin(x) / \cos(x)$?

Solution (continued)

(b)



In this case, exact values are not visible, but rough estimates from the diagram are $\sin(-50^\circ) \approx -0.8$, $\cos(-50^\circ) \approx 0.6$, and $\tan(-50^\circ) \approx -1.2$.

Basic Exercise

Determine the values of the following expressions based on the unit circle:

- | | | | |
|-----------------|--|------------------|--|
| (a) $\cos(0)$ | (b) $\cos\left(\frac{\pi}{2}\right)$ | (c) $\cos(3\pi)$ | (d) $\sin\left(\frac{\pi}{2}\right)$ |
| (e) $\sin(\pi)$ | (f) $\sin\left(-\frac{5\pi}{2}\right)$ | (g) $\tan(0)$ | (h) $\tan\left(\frac{\pi}{2}\right)$. |

Basic Exercise

Estimate the sine, cosine, and tangent of the angle 30° using the unit circle.

A certain angle always corresponds to a right triangle on the unit circle, where the sides are the sine and cosine of the angle. Generally, these cannot be determined using this triangle because the resulting equation has two unknowns, but in certain special cases, this can be done.

Basic Exercise

One angle of a right triangle is 45° and the length of the hypotenuse is 1. Determine the lengths of the legs of the triangle.

Exercise

Determine the values of the following expressions based on the unit circle and a suitable triangle:

$$(a) \cos\left(\frac{\pi}{4}\right) \quad (b) \sin\left(-\frac{3\pi}{4}\right) \quad (c) \tan\left(-\frac{\pi}{4}\right) \quad (d) \tan\left(\frac{3\pi}{4}\right).$$

Exercise

Determine the values of the following expressions based on the unit circle and a suitable triangle:

(a) $\sin\left(\frac{\pi}{6}\right)$

(b) $\sin\left(\frac{\pi}{3}\right)$

(c) $\cos\left(\frac{\pi}{6}\right)$

(d) $\cos\left(\frac{\pi}{3}\right)$

(e) $\tan\left(\frac{\pi}{6}\right)$

(f) $\tan\left(\frac{\pi}{3}\right)$.

Next, we will address the reverse reasoning, i.e., determine the angles corresponding to the values of trigonometric functions using the unit circle.

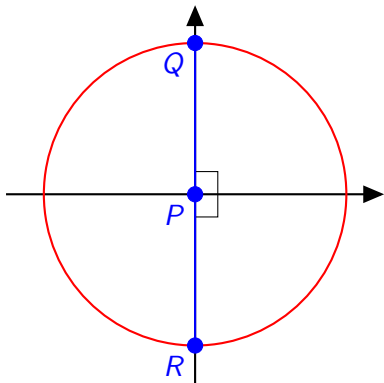
Example

Determine approximately all angles corresponding to the following values based on the unit circle:

(a) the cosine value 0 (b) the sine value $\frac{1}{\sqrt{2}}$ (c) the tangent value $\frac{1}{\sqrt{3}}$.

Solution

- (a) Start by drawing a diagram. First, mark the point P on the x -axis corresponding to a cosine value of 0, and then mark the corresponding points on the circumference of the unit circle, Q and R . Finally, indicate the angles corresponding to these points on the circumference.



Solution (continued)

From the diagram, the point Q corresponds to the angle $\frac{\pi}{2}$. In the interval $(-\pi, \pi]$, there is also another angle whose cosine value is 0, which is the angle $-\frac{\pi}{2}$, corresponding to the point R . Additionally, all 2π -multiples must be considered, which are

$$\dots, -\frac{7}{2}\pi, -\frac{3}{2}\pi, \frac{1}{2}\pi, \frac{5}{2}\pi, \dots$$

and

$$\dots, -\frac{5}{2}\pi, -\frac{1}{2}\pi, \frac{3}{2}\pi, \frac{7}{2}\pi, \dots$$

No other angles need to be found.

The answer from the upper row can be simplified, as all angles are obtained by adding all possible 2π -integer multiples to the angle $\frac{\pi}{2}$. By adding 2π for instance -2 times, we get the angle $\frac{\pi}{2} - 2 \cdot 2\pi = -\frac{7}{2}\pi$. All these 2π -additions can be expressed with the coefficient n , $n \in \mathbb{Z}$, which represents any integer. Thus, the upper row can be written as

$$\frac{\pi}{2} + 2\pi n, \quad n \in \mathbb{Z}.$$

For example, the angle $-\frac{7}{2}\pi$ corresponds to $n = -2$.

Solution (continued)

The lower row can be similarly simplified to

$$-\frac{\pi}{2} + 2\pi n, \quad n \in \mathbb{Z}.$$

These can be combined into

$$\pm \frac{\pi}{2} + 2\pi n, \quad n \in \mathbb{Z}.$$

This is not accidental, as with the cosine equation $\cos(x) = a$, due to symmetry, all solutions are always in the form

$$x = \pm \text{solution in the interval } [0, \pi] + 2\pi n, \quad n \in \mathbb{Z}.$$

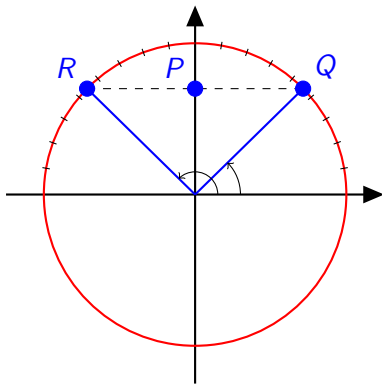
To solve such an equation, it is sufficient to find one solution, after which all solutions can be written in the above form using symmetry. In this example, all solutions are actually spaced half a turn apart, which allows the solution $\pm \frac{\pi}{2} + 2\pi n$, $n \in \mathbb{Z}$, to be further simplified to

$$\frac{\pi}{2} + \pi n, \quad n \in \mathbb{Z}.$$

For instance, with $n = -3$, we get the angle $-\frac{5}{2}\pi$. Generally, such further simplification is not possible with cosine, and the general solution retains two angles in the interval $(-\pi, \pi]$ (or the interval $[0, 2\pi)$) and their 2π -multiples.

Solution (continued)

- (b) First, mark the point P on the y -axis corresponding to a sine value of 0.7, and then mark the corresponding points on the circumference of the unit circle, Q and R . Finally, indicate the angles corresponding to these points on the circumference. The dashed lines illustrate how the points Q and R are determined based on the point P .



Solution (continued)

Based on the diagram, there are two angles in the interval $(-\pi, \pi]$, approximately $45^\circ = \frac{\pi}{4}$ and $135^\circ = \frac{3}{4}\pi$. (In fact, these are the exact values). Additionally, all 2π -multiples must be considered. This time, all answers cannot be given in one expression, but it is possible with two expressions. The answer is thus all angles of the form

$$\frac{\pi}{4} + 2\pi n \quad \text{or} \quad \frac{3}{4}\pi + 2\pi n, \quad n \in \mathbb{Z}.$$

Note that the other angle in the interval $(-\pi, \pi]$ (or $[0, 2\pi)$) is obtained by subtracting the other angle from half a circle based on symmetry. This is not accidental, as with the sine equation $\sin(x) = a$, due to symmetry, all solutions are always in the form

$$x = \text{solution in the interval } \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] + 2\pi n$$

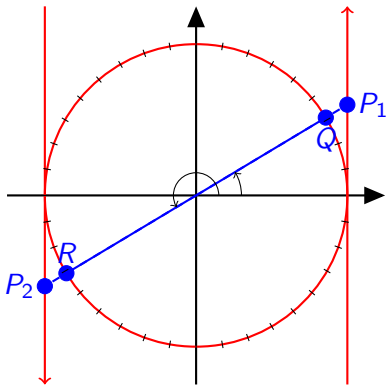
or

$$x = \pi - \text{solution in the interval } \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] + 2\pi n, \quad n \in \mathbb{Z}.$$

To solve such an equation, it is sufficient to find one solution, after which all solutions can be written in the above form using symmetry.

Solution (continued)

- (c) First, mark the points P_1 and P_2 on the tangent lines corresponding to a tangent value of 0.6, and then mark the corresponding points on the circumference of the unit circle, Q and R . Finally, indicate the angles corresponding to these points on the circumference.



Solution (continued)

Based on the diagram, there are two angles in the interval $(-\pi, \pi]$, approximately $\frac{1}{6}\pi$ and $\frac{7}{6}\pi$. Additionally, all 2π -multiples must be considered. Since the angles in the interval $(-\pi, \pi]$ are spaced by π , all angles are spaced by π , so the answer can be expressed in one formula. The answer is thus

$$\frac{1}{6}\pi + \pi n, \quad n \in \mathbb{Z}.$$

All solutions to the equation of the form $\tan(x) = a$ can be inferred from the corresponding cosine and sine equations similarly using one solution and the regularity of solutions. The exact form of this solution in the tangent case is left for the reader to consider.

Basic Exercise

Determine approximately all angles corresponding to the following values based on the unit circle:

- | | | |
|------------------------------|----------------------|----------------------------------|
| (a) the cosine value
-0.5 | (b) the sine value 1 | (c) the tangent value
-0.3 |
| (d) the cosine value π | (e) the sine value 2 | (f) the tangent value
π . |

Basic Exercise

Deduce the solutions to the following problems based on the unit circle:

- (a) The sine of the angle $\frac{2}{3}\pi$ is $\frac{\sqrt{3}}{2}$. What other angles in the interval $(-\pi, \pi]$ have the same sine value? Provide all angles corresponding to the same sine value.
- (b) The cosine of the angle $\frac{5}{9}\pi$ is approximately -0.2 . What other angles in the interval $(-\pi, \pi]$ have approximately the same cosine value? Provide all angles corresponding to approximately the same cosine value.
- (c) The tangent of the angle $\frac{13}{18}\pi$ is approximately -1.2 . What other angles in the interval $(-\pi, \pi]$ have approximately the same tangent value? Provide all angles corresponding to approximately the same tangent value.

Note that in the previous example, part (a) solved the equation $\cos(x) = 0$, part (b) (approximately) solved the equation $\sin(x) = 0.5$, and part (c) (approximately) solved the equation $\tan(x) = 0.3$. This is exactly the technique for solving trigonometric equations that is recommended.

Usually, one does not see the exact solution from the unit circle but refers to a table or uses a calculator. This will be revisited in section 1.5.3.9, where solving this type of trigonometric equations will be discussed in more detail. The essential part of the solution technique is included in the previous example and problem. If this concept is clear now, solving trigonometric equations will be easy later; if it is not yet clear, you will need to work through it later in section 1.5.3.9.

However, let's discuss the graphs of trigonometric functions, trigonometric formulas, and the inverse functions of trigonometric functions before delving deeper into equation solving.

Graphs of Trigonometric Functions

The graphs of the sine and cosine functions are known to be waves. Let's see how they relate to the unit circle.

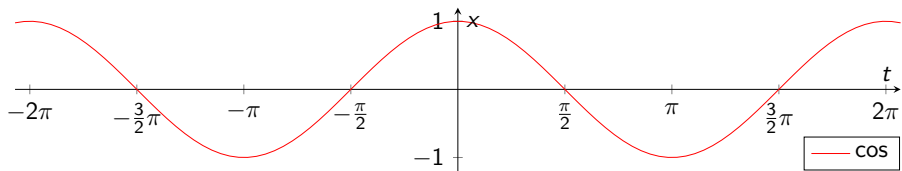
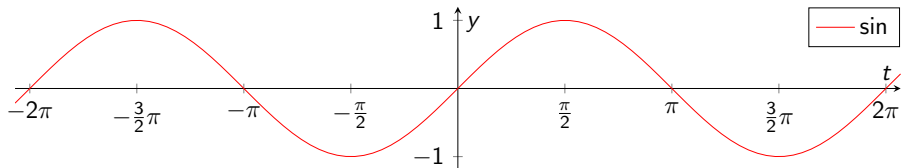
From the unit circle, we can determine, for example, the values for sine:

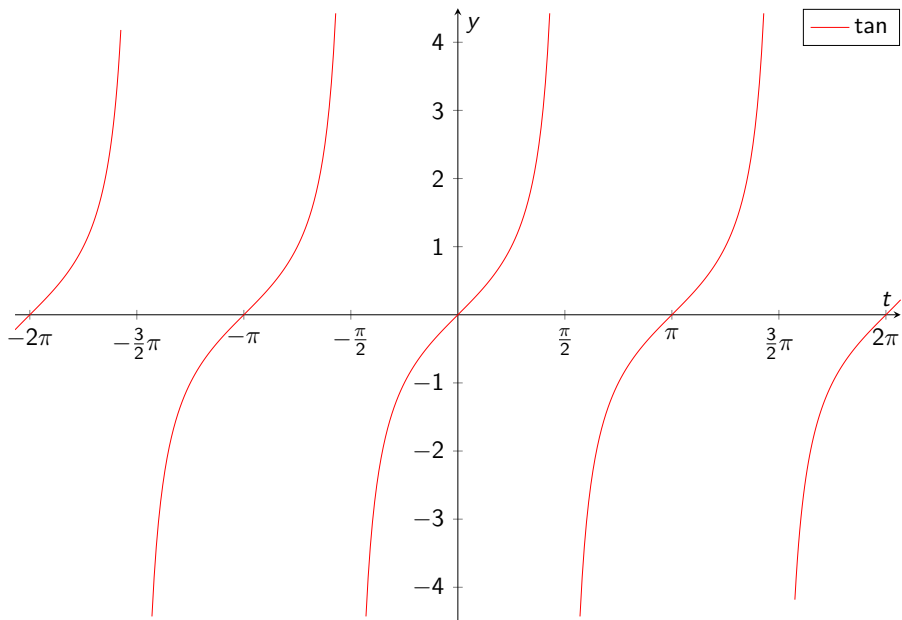
$$\sin\left(-\frac{\pi}{2}\right) = -1, \quad \sin(0) = 0, \quad \sin\left(\frac{\pi}{2}\right) = 1 \quad \text{and} \quad \sin(-\pi) = 0,$$

from which we can deduce that the graph of sine passes through the points $(-\frac{\pi}{2}, -1)$, $(0, 0)$, $(\frac{\pi}{2}, 1)$, and $(-\pi, -1)$. The entire graph is formed from all such points. This is well illustrated in the simulation at [this link](#).

In the previous points, it is essential to note that their horizontal coordinate is not the x-coordinate of the unit circle point but rather the angle of the unit circle. Thus, in the coordinate system of the sine graph, the horizontal axis is the angle t and not x . However, the vertical axis remains the same as in the unit circle. The same applies to the tangent graph. In the cosine graph, the horizontal axis is the angle, while the vertical axis is the horizontal axis of the unit circle.

The graphs of sine, cosine, and tangent functions are shown in the following images.





It is essential to be able to see the following from the graphs:

- the periodicity of the functions
- the lengths of the periods of the functions
- the domains of the functions
- the ranges of the functions
- the possible symmetry of the functions.

Additionally, it can be observed from the graphs that for every $x \in \mathbb{R}$:

$$\sin\left(x + \frac{\pi}{2}\right) = \cos(x),$$

which can be expressed verbally as the sine phase, or the value of its inner function, is ahead of the cosine by an angle of $\frac{\pi}{2}$. Therefore, cosine can be replaced with the sine function and vice versa.

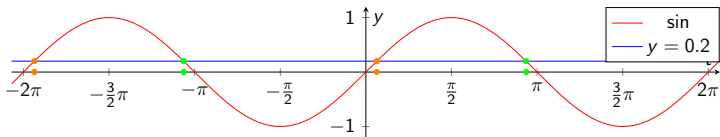
Let's also review what it means to solve angles corresponding to values of trigonometric functions using the graph.

Example

Determine, based on the sine graph, all angles that correspond to a sine value of 0.2.

Solution

Start by drawing the sine graph. We are interested in the horizontal axis values where the graph reaches a height of 0.2. Therefore, add the line $y = 0.2$ to the graph and determine the x-coordinates of the intersection points. The intersection points and their corresponding x-values are shown in the image with orange and green points. The colors highlight which angles are multiples of 2π from each other.



Based on the image, the angles are *approximately* $\frac{1}{18}\pi$ and $\frac{17}{18}\pi$ and their multiples, i.e., angles

$$\frac{1}{18}\pi + 2\pi n \quad \text{or} \quad \frac{17}{18}\pi + 2\pi n, \quad n \in \mathbb{Z}.$$

Trigonometric Formulas

From the definitions of sine and cosine, the following properties follow:

- (1) For every $t \in \mathbb{R}$, we have

$$\sin^2(t) + \cos^2(t) = 1,$$

where $\sin^2(x) = (\sin(x))^2$ and $\cos^2(x) = (\cos(x))^2$.

- (2) Sine and cosine are 2π -periodic, meaning for every $t \in \mathbb{R}$ we have

$$\sin(t + 2\pi) = \sin(t) \quad \text{and} \quad \cos(t + 2\pi) = \cos(t),$$

and also

$$\sin(t + n2\pi) = \sin(t) \quad \text{and} \quad \cos(t + n2\pi) = \cos(t), \quad n \in \mathbb{Z}.$$

- (3) Sine is an odd function and cosine is an even function, meaning for every $t \in \mathbb{R}$ we have

$$\sin(-t) = -\sin(t) \quad \text{and} \quad \cos(-t) = \cos(t).$$

- (4) The range of sine is $\mathcal{R}(\sin) = [-1, 1]$ and the range of cosine is $\mathcal{R}(\cos) = [-1, 1]$. (What are their domains?)

You should be able to know these properties. In addition to memorization, it is important to be able to deduce them based on the unit circle.

Property (1) holds for any inner function, and this property can sometimes simplify the expression being handled nicely. For example, we have

$$\cos^2(2t^3 - 4) + \sin^2(2t^3 - 4) = 1.$$

The equation $\sin(t + 2\pi n) = \sin(t)$ tells us that the sine graph is always at the same height every 2π multiples apart. On the other hand, it indicates that if t is a solution to the sine equation, then any number $t + 2\pi n$, $n \in \mathbb{Z}$, is also a solution to the same equation. The cosine equation has the same interpretations.

The equation $\cos(-t) = \cos(t)$ indicates that the cosine graph is symmetric with respect to the y -axis. On the other hand, it means that if t is a solution to the cosine equation, then $-t$ is also a solution to the same equation. The equation $\sin(-t) = -\sin(t)$ indicates that the sine graph is symmetric with respect to the origin (point). Unlike cosine, it does not indicate that if t is a solution to the sine equation, then $\pi - t$ is also a solution, but this fact is indicated by the equation $\sin(\pi - t) = \sin(t)$.

Sine and cosine are defined for all values of the variable $t \in \mathbb{R}$.

Basic Exercise

Determine based on the unit circle the properties of the tangent function:

- | | |
|------------|-----------------------|
| (a) period | (b) domain |
| (c) range | (d) evenness/oddness. |

In addition to the above, trigonometric functions have many other formulas. Here, we will go through examples only for sine. The corresponding formulas for cosine and tangent can be found in handbooks. It is not necessary to memorize these formulas, and their derivation is not as straightforward as for the formulas discussed above. However, it is important to be aware of their existence so that you can find them if needed. As with other functions, **trigonometric functions should not use formulas that seem natural but only those found in handbooks.**

Example

Using a counterexample, show that

$$\sin(x + y) \neq \sin(x) + \sin(y),$$

for some $x, y \in \mathbb{R}$.

For sine, we have the sum formulas

$$\sin(x \pm y) = \sin(x) \cos(y) \pm \cos(x) \sin(y), \quad (8)$$

$$\sin(x) + \sin(y) = 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right),$$

$$\sin(x) - \sin(y) = 2 \sin\left(\frac{x-y}{2}\right) \cos\left(\frac{x+y}{2}\right),$$

$$\cos(x) \pm \sin(x) = \sqrt{2} \sin\left(\frac{\pi}{4} \pm x\right).$$

From the formula (8), we can straightforwardly derive double-angle formulas, such as

$$\sin(2x) = 2 \sin(x) \cos(x).$$

Using these formulas, other similar formulas can also be derived, such as the triple-angle formula

$$\sin(3x) = 3 \sin(x) - 4 \sin^3(x)$$

and the half-angle formula

$$\sin\left(\frac{x}{2}\right) = \pm \sqrt{\frac{1 - \cos(x)}{2}}.$$

Based on the above formulas and the formula $\cos^2(x) + \sin^2(x) = 1$, we can further derive power formulas for trigonometric functions, such as

$$\sin^2(x) = \frac{1}{2}(1 - \cos(2x)),$$

$$\sin^3(x) = \frac{1}{4}(3 \sin(x) - \sin(3x)),$$

and

$$\sin^4(x) = \frac{1}{8}(\cos(4x) - 4 \cos(2x) + 3).$$

Power formulas are useful, for example, when integrating trigonometric functions.

The following formula is also useful:

$$\cos(x \pm y) = \cos(x) \cos(y) \mp \sin(x) \sin(y).$$

Inverse Functions of Trigonometric Functions

For the angle π , we have $\cos(\pi) = -1$. Conversely, it holds that $\arccos(-1) = \pi$. Here, $\arccos$ is the inverse function of cosine, called the arccosine or arcus cosine. It is also denoted as $\cos^{-1}$, as commonly used in calculators due to space constraints, but this notation can easily be confused with the reciprocal[†]

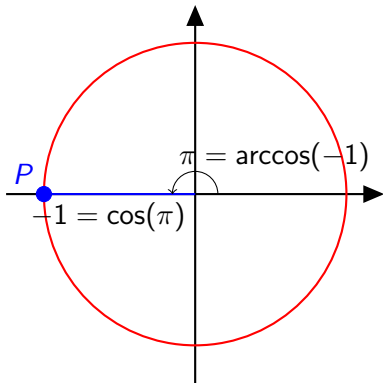
$$(\cos(x))^{-1} = \frac{1}{\cos(x)},$$

because for trigonometric functions, the notation $\cos^n(x)$ usually means the same as $(\cos(x))^n$. Therefore, in this and later courses, we prefer the notation $\arccos$ for the arcus cosine. This also helps to understand why the corresponding command in MATLAB is called `acos`. Similarly, the inverse functions for sine and tangent are denoted by $\arcsin$ and $\arctan$.

[†]The reciprocal and the inverse function are different concepts. The reciprocal is a number that cancels out the effect of multiplication, while the inverse function cancels out the effect of a function. For example, the reciprocal of cosine is $\cos(x)(\cos(x))^{-1} = 1$ and the inverse function of cosine (simplified) is $\cos(\arccos(x)) = x = \arccos(\cos(x))$.

There are many details associated with inverse functions, which will be examined more closely later in Section 1.7. At this stage, the topic is covered only enough to understand what happens precisely when solving a trigonometric equation. Practically, this means internalizing that the inverse function always provides only one solution to a trigonometric equation, and the remaining solutions are inferred based on the regularity of the solutions.

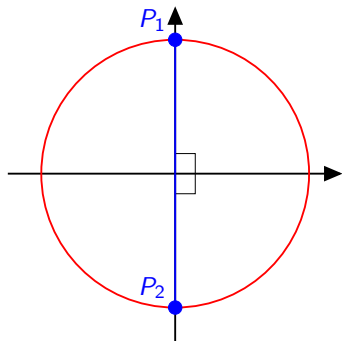
Cosine returns the **horizontal coordinate** of the unit circle's **perimeter point** corresponding to the given angle x , $\cos(x)$. Its inverse function works in reverse, i.e., it returns the **angle** corresponding to the given **horizontal coordinate** of a unit circle's **perimeter point**. This is illustrated in the following image for the case of the angle π .



However, there is an issue with the description above, which becomes apparent in, for example, the image corresponding to the angle $\frac{\pi}{2}$. The issue is that the cosine value for each angle is unique, whereas each horizontal coordinate corresponds to infinitely many angles.

For example, the horizontal coordinate 0 corresponds to angles $\frac{\pi}{2}$ and $-\frac{\pi}{2}$ and all their 2π multiples. Since the arc cosine is a function, its value must be unique, meaning it always returns only one angle. But which one? The arc cosine is defined to return an angle in the interval $[0, \pi]$. You can see this concretely by using a calculator to compute the arc cosine of any number in the interval $[-1, 1]$: the calculator returns only one angle, which is always from the aforementioned interval. **Thus, the arc cosine does not provide all solutions to the cosine equation but only one of them.**

Similarly, the arc sine always returns an angle in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ and the tangent returns an angle in the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$.



If the inverse function provides only one angle, how then are the remaining angles found?

Let's use an analogy. How do you find the solution to the equation $x^2 = 4$? The inverse function of the square (simplified) is the square root. It provides only one solution, $\sqrt{2}$. The remaining solutions, $-\sqrt{2}$, are found by adding them, based on the symmetry of the solutions (with respect to the y -axis).

Similarly, for trigonometric functions, the inverse function gives one solution, and the remaining solutions are added based on the regularity of the solutions. For example, one solution to the equation $\cos(x) = 0$ is $x = \arccos(0) = \frac{\pi}{2}$ (found using the unit circle, table, or calculator). From there, you can deduce based on the unit circle that $x = -\arccos(x) = -\frac{\pi}{2}$ is another solution, and further, based on periodicity, that all 2π multiples are also solutions. Thus, it holds that

$$\begin{aligned}\cos(x) &= 0 \\ x &= \pm \frac{\pi}{2} + 2\pi n, \quad n \in \mathbb{Z}.\end{aligned}$$

Finally, here are the domains and ranges of trigonometric inverse functions and their graphs:

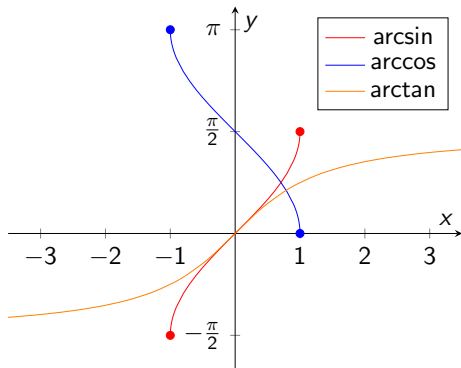
$$\arcsin: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], x \mapsto \arcsin(x),$$

$$\arccos: [-1, 1] \rightarrow [0, \pi], x \mapsto \arccos(x),$$

$$\arctan: \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), x \mapsto \arctan(x).$$

(The notation $x \mapsto x^2$ corresponds to $f(x) = x^2$, which indicates how the function transforms the value of the variable into the function's value.

However, this notation is somewhat awkward when the function has a well-known name, as both sides of the equation are then the same. For example, for the arc sine, the notation would be $\arcsin(x) = \arcsin(x)$. Therefore, an alternative and less commonly used notation was used above.)



Based on the graphs, it appears that the arcsine and arctangent functions are odd functions. This is indeed the case, meaning that

$$\arcsin(-x) = -\arcsin(x)$$

and

$$\arctan(-x) = -\arctan(x).$$

Based on this, one can write, for example,

$\pi - \arcsin(-0.1) = \pi + \arcsin(0.1)$. However, the arccosine is not an even function, even though the cosine function itself is even. Therefore, for example, $\arccos(-1) \neq \arccos(1)$.

Basic Exercise

Let $n \in \mathbb{Z}$. Which of the following equations are true:

(a) $\cos(0) = 1$

(b) $\cos(2\pi) = 1$

(c) $\cos(-8\pi) = 1$

(d) $\cos(n2\pi) = 1$

(e) $\arccos(1) = 0$

(f) $\arccos(1) = 2\pi$

(g) $\arccos(1) = -8\pi$

(h) $\arccos(1) = n2\pi?$

Solving Trigonometric Equations 1

In this chapter, we examine trigonometric equations that can be simplified to the forms

$$\sin(x) = a, \quad \cos(x) = a \quad \text{and} \quad \tan(x) = a,$$

where $a \in \mathbb{R}$. For example, the equation $3 \sin(x) + 2 = 6 \sin(x) + 1$ falls into this category because it can be rewritten as $\sin(x) = \frac{1}{3}$. What is crucial in these equations is that there is only one trigonometric function, and its inner function is $f(x) = x$. More complex cases will be considered in later chapters, but solving them ultimately reduces to the cases discussed in this chapter.

Let's start with an example.

Example

Solve the following equations:

$$(a) \cos(x) = 0 \qquad (b) \sin(x) = \frac{1}{\sqrt{2}} \qquad (c) \tan(x) = 0.6.$$

Solution

- (a) **Method 1.** We need to find all angles x that satisfy the equation $\cos(x) = 0$. We already solved this problem using the unit circle, which is indeed the most convenient method for solving this equation. From the unit circle (draw how), it can be seen that in the interval $[0, \pi]$ the solution is $x = \frac{\pi}{2}$, and then, by symmetry or checking the unit circle, it follows that in the interval $(-\pi, 0)$ the solution is $x = -\frac{\pi}{2}$. There are no other solutions in the interval $(-\pi, \pi]$, and solutions in other intervals are obtained by adding multiples of 2π . Thus, the solutions to the equation $\cos(x) = 0$ are

$$x = \pm \frac{\pi}{2} + 2\pi n, \quad n \in \mathbb{Z}.$$

The previous solution can be simplified further to

$$x = \frac{\pi}{2} + \pi n, \quad n \in \mathbb{Z},$$

as mentioned in an earlier example.

Solution (continued)

Method 2. The same equation can also be solved using the inverse function of cosine. Thus, it follows that the solution to $\cos(x) = 0$ can be written as

$$x = \pm \arccos(0) + 2\pi n, \quad n \in \mathbb{Z}.$$

In this case, the solution in the interval $[0, \pi]$ is $x = \arccos(0) = \frac{\pi}{2}$, so the remaining solutions are determined based on the unit circle as in the previous solution:

$$x = \pm \frac{\pi}{2} + 2\pi n, \quad n \in \mathbb{Z}.$$

The examined equation can also be solved using a table, from which it can be seen that in the interval $[0, \pi]$ the solution is $x = \frac{\pi}{2}$ and in the interval $(\pi, 2\pi)$ the solution is $\frac{3}{2}\pi$, either of which is sufficient for writing the solution. The remaining solutions are determined based on the unit circle.

Solution (continued)

(b) We need to find all angles x that satisfy the equation $\sin(x) = \frac{1}{\sqrt{2}}$.

We previously solved a version of this equation with an approximate value on the right side was solved. Now, however, we want the exact value, so we cannot just make an estimate.

The exact value can be written using the inverse function of sine, so in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ the solution is $x = \arcsin\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$. From this, it can further be concluded directly by symmetry or verified using the unit circle that in the interval $(-\pi, \pi]$ there is also the solution $x = \pi - \arcsin\left(\frac{1}{\sqrt{2}}\right) = \frac{3}{4}\pi$ and that there are no other solutions in this interval. Additionally, based on the unit circle, solutions that are multiples of 2π are also valid, so the solutions to the equation $\sin(x) = \frac{1}{\sqrt{2}}$ are

$$x = \arcsin\left(\frac{1}{\sqrt{2}}\right) + 2\pi n \quad \text{or} \quad x = \pi - \arcsin\left(\frac{1}{\sqrt{2}}\right) + 2\pi n, \quad n \in \mathbb{Z}$$
$$x = \frac{\pi}{4} + 2\pi n \quad \text{or} \quad x = \frac{3}{4}\pi + 2\pi n, \quad n \in \mathbb{Z}.$$

Solution

The examined equation can also be solved using a table, from which it can be seen that in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ the solution is $x = \frac{\pi}{4}$ and in the interval $(\frac{\pi}{2}, \frac{3}{2}\pi)$ the solution is $\frac{3}{4}\pi$, either of which is sufficient for writing the solution. The remaining solutions are determined based on the unit circle.

- (c) The solution to the equation $\tan(x) = 0.6$ follows similarly to the previous cases, namely by determining one exact solution and then finding other solutions based on periodicity, but since the exact value of the one solution cannot be obtained, it must be left written in terms of the inverse function of the tangent:

$$\begin{aligned}\tan(x) &= \frac{3}{5} \\ x &= \arctan\left(\frac{3}{5}\right) + \pi n, \quad n \in \mathbb{Z}.\end{aligned}$$

Thus, the problem cannot be solved exactly using the unit circle or a table. Compare this to the solution of $x^2 = 7$, which simplifies to $x = \pm\sqrt{7}$ and does not further simplify, thus it must be expressed in terms of the inverse function, i.e., the square root.

In this chapter, solving trigonometric equations is divided into two steps, similar to the previous example:

- 1 First, find one (any) solution of the equation. This is always possible using the inverse function, and in some cases with the help of the unit circle or a table.
- 2 Based on the solution from the previous step, write all solutions of the equation according to the regularities that can be inferred from the unit circle.

Basic Exercise

Solve the following equations:

(a) $\sin(x) = 1$

(b) $2(5 \cos(x) - 1) = 1$

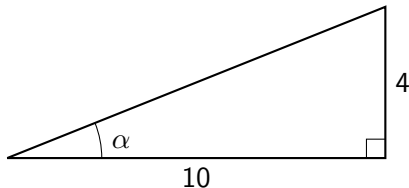
(c) $2 \tan(x) + 6 = \tan(x) + 5.$

Example

The sides of a right triangle are 10 and 4. Find the value of the smallest angle.

Solution

Let's start by drawing a diagram.



For the angle α , we have

$$\tan(\alpha) = \frac{4}{10}$$

$$\alpha = \arctan\left(\frac{2}{5}\right) + \pi n, \quad n \in \mathbb{Z}.$$

But which of these angles is the correct one?

Solution (continued)

This could be tested by calculating angles for different values of the coefficient n , but since we are interested in the angle in the interval $(0^\circ, 90^\circ)$, and because the inverse function of the tangent returns the solution precisely from this interval, and since there is only one angle in this interval for the tangent equation of the form $\tan(\alpha) = a$, we already know without calculating that the sought angle is $\alpha = \arctan\left(\frac{2}{5}\right)$. Using a calculator, the approximate value is $\alpha \approx 21.8^\circ$.

It is important to note that the tangent equation solved above always has infinitely many solutions, but only one of these solutions fits the problem solved here. It should also be repeated that for a right triangle, trigonometric inverse functions always return the desired angle in the interval $(0^\circ, 90^\circ)$. The fact that solving trigonometric equations in middle school was done without mentioning the infinite number of solutions was because the situation was limited to right triangles.

It is advisable to write the solutions of a trigonometric equation either entirely in radians or entirely in degrees.

Basic Exercise

Find all solutions to the equation $\cos(x) = -1$.

Solving Trigonometric Equations 2

In this chapter, we consider trigonometric equations that can be simplified to the forms

$$\sin(f(x)) = a, \quad \cos(f(x)) = a \quad \text{and} \quad \tan(f(x)) = a,$$

where $a \in \mathbb{R}$. The key aspect of these equations is that they involve only one trigonometric function, but unlike in Chapter 1.5.3.9, the inner function can be more complex. Such equations are solved in two steps:

- 1 First, let $f(x) = \alpha$ and introduce a new unknown for the inner function. This transforms the equation into

$$\sin(\alpha) = a, \quad \cos(\alpha) = a \quad \text{or} \quad \tan(\alpha) = a,$$

which can be solved using the methods presented in Chapter 1.5.3.9.

- 2 After solving for the unknown α , solve for x from the equation $f(x) = \alpha$.

Changing the variable to $f(x) = \alpha$ is not mandatory; one could first solve the equation with respect to the inner function $f(x)$ and then solve the resulting equation for x . However, using the auxiliary variable α can clarify the intermediate steps.

Example

Solve the equation

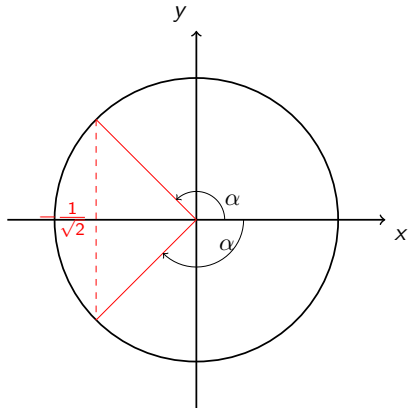
$$\cos\left(x^2 - \frac{\pi}{2}\right) = -\frac{1}{\sqrt{2}}.$$

Solution

Start by letting $x^2 - \frac{\pi}{2} = \alpha$, which simplifies the problem to solving the simpler equation $\cos(\alpha) = -\frac{1}{\sqrt{2}}$. By using the unit circle and a table, it can be found that the solution to the equation in the interval $[0, \pi]$ is^a $\frac{3\pi}{4}$, so the general solution is

$$\alpha = \pm \frac{3\pi}{4} + 2\pi n, \quad n \in \mathbb{Z}.$$

^aOne could also remember that the value $\frac{1}{\sqrt{2}}$ corresponds to a 45° angle, so the upper angle of the figure is thus $90^\circ + 45^\circ$.



Solution (continued)

We then need to solve for x . We have

$$x^2 - \frac{\pi}{2} = \alpha$$

$$x^2 - \frac{\pi}{2} = \pm \frac{3\pi}{4} + 2\pi n, \quad n \in \mathbb{Z}.$$

From here, the solution divides into two cases depending on whether the right-hand side of the equation has $+$ or $-$.

In the $+$ case, we get

$$x^2 = \frac{5\pi}{4} + 2\pi n, \quad n \in \mathbb{Z}.$$

Since $x^2 > 0$, only values $n = 0, 1, 2, \dots$ are valid, giving the solutions

$$x = \pm \sqrt{\frac{5\pi}{4} + 2\pi n}, \quad n \in \mathbb{Z}_{\geq 0}.$$

Solution (continued)

In the – case, we have

$$x^2 = -\frac{\pi}{4} + 2\pi m, \quad m \in \mathbb{Z}.$$

Since $x^2 > 0$, only values $m = 1, 2, \dots$ are valid. However, by choosing the angle $\frac{7\pi}{4}$ instead of $-\frac{\pi}{4}$ (i.e., performing the change of variable $m = n + 1$), the solution can be written using the same coefficient $n \in \mathbb{Z}_{\geq 0}$ as in the previous solution:

$$x = \pm \sqrt{\frac{7\pi}{4} + 2\pi n}, \quad n \in \mathbb{Z}_{\geq 0}.$$

Thus, the solution to the equation is

$$x = \pm \sqrt{\frac{5\pi}{4} + 2\pi n} \quad \text{or} \quad x = \pm \sqrt{\frac{7\pi}{4} + 2\pi n}, \quad n \in \mathbb{Z}_{\geq 0}.$$

The manipulation with the latter solution is not necessary, but it allows the final solution to be written using the coefficient n . Otherwise, the latter equation would have needed to keep m or replace it with $n + 1$.

Solution (continued)

It is worth noting that if the auxiliary variable α had not been used, one could have written directly

$$\cos\left(x^2 - \frac{\pi}{2}\right) = -\frac{1}{\sqrt{2}}$$
$$x^2 - \frac{\pi}{2} = \pm \frac{3\pi}{4} + 2\pi n, \quad n \in \mathbb{Z},$$

and then proceeded as above.

Basic Exercise

Solve the equation $2 \sin(3x + 1) = \sqrt{2}$.

Exercise

The diameter of the London Eye Ferris wheel is 60 m, and one complete rotation takes about half an hour. Form a function that describes a passenger's height, assuming that at time $t = 0$ their carriage is at the bottom position, that the carriage moves at a constant speed throughout the rotation, and that the zero level of height is chosen (a) at the axis of the Ferris wheel (b) at the bottom position of the carriage.

Solving Trigonometric Inequalities

Solving trigonometric inequalities is easier when using the unit circle instead of a sign chart. This is because the unit circle graphically represents the problem, making it somewhat like solving the inequality using the graph of the function. When using the unit circle, the solution algorithm takes the following form:

- 1 Write the given inequality in the form $f(x) > a$, $f(x) < a$, $f(x) \geq a$, or $f(x) \leq a$, where a is a real number and f is a trigonometric function.
- 2 Find the zero points and discontinuities of the equation $f(x) = a$.
- 3 Mark the angles corresponding to the zero points and discontinuities on the unit circle.
- 4 Determine, based on the trigonometric function, which intervals of the unit circle satisfy the original inequality.
- 5 Based on the marked intervals, first write a solution for one period (e.g., $[0, 2\pi)$) and then add the periods of the solutions (and simplify the answer if possible).

Example

Solve the inequality $\tan(x) > 1$.

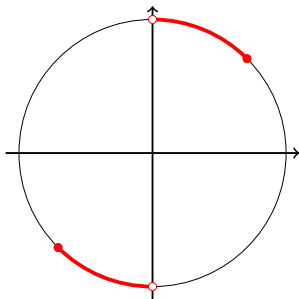
Solution

Zero points of $\tan(x) = 1$:

$$x = \frac{\pi}{4} + \pi n, \quad n \in \mathbb{Z}.$$

Discontinuities: $\frac{\pi}{2} + n\pi, n \in \mathbb{Z}$.

Mark the angles corresponding to the zero points and discontinuities on the unit circle. Since the period of the tangent function is π and the solution in the interval $[0, \pi)$ is $(\frac{\pi}{4}, \frac{\pi}{2})$, the general solution is $(\frac{\pi}{4} + \pi n, \frac{\pi}{2} + \pi n), n \in \mathbb{Z}$.



The endpoints of the solution interval (a, b) must be chosen sensibly, as discussed in the following example.

Example

Solve the following inequalities:

$$(a) \cos(x) > 0$$

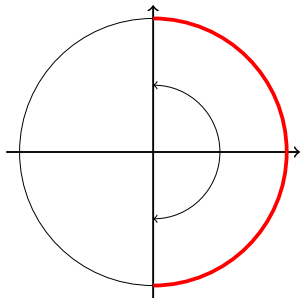
$$(b) \sin(t) < -0.9.$$

Solution

- (a) Let's begin again by illustrating the situation with the unit circle. For clarity, we use degrees instead of radians.

From the diagram, it is clear that the solution is the interval $(-90^\circ, 90^\circ)$ along with its 360° multiples, i.e.,

$$(-90^\circ + 360^\circ n, 90^\circ + 360^\circ n), \quad n \in \mathbb{Z}.$$

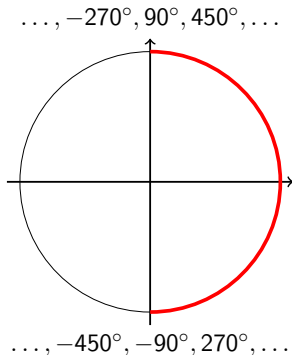


Solution

To further clarify, each endpoint of an angular interval on the unit circle corresponds to infinitely many angle values. This is illustrated in the image below.

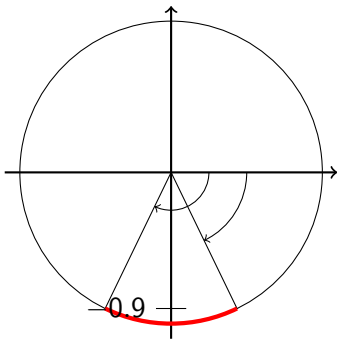
The endpoints of the interval cannot be arbitrarily chosen from these options, as different choices correspond to different intervals.

For example, choosing $(270^\circ, 90^\circ)$ is incorrect because it results in an empty set, that is, an interval containing no elements. Valid options include $(-450^\circ, -270^\circ)$, $(-90^\circ, 90^\circ)$, and $(270^\circ, 450^\circ)$, as they correspond to the correct arc. One must choose a lower bound and the corresponding **next** upper bound.



Solution (continued)

- (b) Let's start by visualizing the situation where $\sin(t) < -0.9$ on the unit circle.



Let's solve for the angles corresponding to the endpoints:

$$\sin(t) = -0.9$$

$$t = \arcsin(-0.9) + 2\pi n \quad \text{or} \quad t = \pi - \arcsin(-0.9) + 2\pi n, \quad n \in \mathbb{Z}.$$

Solution (continued)

The angle $\arcsin(-0.9)$ lies in the interval $(-\frac{\pi}{2}, 0)$, based on the range of the inverse sine function $([-\frac{\pi}{2}, \frac{\pi}{2}])$. The angle $\pi - \arcsin(-0.9)$ lies in $(\pi, \frac{3\pi}{2})$ and is not valid as a lower bound if the upper bound is chosen as $\arcsin(-0.9)$. On the other hand, the angle $-\pi - \arcsin(-0.9)$ lies in $(-\frac{3\pi}{2}, -\pi)$ and is valid as a lower bound. A valid solution interval is, for example,

$$(-\pi - \arcsin(-0.9), \arcsin(-0.9))$$

along with its 2π -periodic repetitions:

$$(-\pi - \arcsin(-0.9) + 2\pi n, \arcsin(-0.9) + 2\pi n), \quad n \in \mathbb{Z}.$$

Another simple option would be

$$(\pi - \arcsin(-0.9) + 2\pi n, 2\pi + \arcsin(-0.9) + 2\pi n), \quad n \in \mathbb{Z}.$$

Note also that using the oddness of arcsine, i.e., the identity $\arcsin(-x) = -\arcsin(x)$, one could simplify as follows:

$$\pi - \arcsin(-0.9) = \pi + \arcsin(0.9).$$

Basic Exercise

Solve the following inequalities:

$$(a) \sin(x) < 1$$

$$(b) \cos(x) > 0.3$$

$$(c) \tan(x) \leq -1.$$

Example

Solve the inequality $\sin(2x) > \frac{1}{2}$.

Solution

Zero points:

$$\sin(2x) = \frac{1}{2}$$

$$2x = \frac{\pi}{6} + 2\pi n \quad \text{or} \quad 2x = \frac{5\pi}{6} + 2\pi n, \quad n \in \mathbb{Z}$$

$$x = \frac{\pi}{12} + \pi n \quad \text{or} \quad x = \frac{5\pi}{12} + \pi n, \quad n \in \mathbb{Z}.$$

Discontinuities: None.

Since the period of the function $\sin(2x)$ is π and the solution in the interval $[0, \pi)$ is $(\frac{\pi}{12}, \frac{5\pi}{12})$, the general solution is $(\frac{\pi}{12} + \pi n, \frac{5\pi}{12} + \pi n)$, $n \in \mathbb{Z}$.

Basic Exercise

Solve the inequality $\cos(\pi x) > 0$.

Solving Trigonometric Equations 3

Finally, let's briefly discuss solving more complex trigonometric equations.

The key idea in solving an equation with multiple trigonometric terms is to try to simplify the problem to solving an equation with only one trigonometric term. This reduces the problem to situations discussed in previous chapters. Trigonometric identities are often needed here. It is difficult to provide a universal guideline, but a few tips can be very helpful. Additionally, experimentation and trial and error are necessary.

If the equation involves only sines and cosines, it is often useful to first try to convert it into a tangent equation.

Basic Exercise

Solve the equation $\sin(x) = \cos(x)$.

If the trigonometric equation contains different types of trigonometric functions, it is advisable to try to convert it using identities into a form where only one type of function appears with the same inner function. This way, it can sometimes be interpreted as a polynomial in the given trigonometric function, which can then be solved in two stages as discussed in Chapter 1.5.1.4. For example, the equation $3\sin^2(x) - 2\sin(x) + 2 = 0$ can first be written as $3y^2 - 2y + 2 = 0$ and then solved for x from $\sin(x) = y$. Also, remember to use the zero product property if applicable.

Example

If it is known that $\alpha \in \left[\frac{3\pi}{2}, \frac{5\pi}{2}\right]$ and $\sin(\alpha) = -0.5$, what is $\cos(\alpha)$?

Solution

$$\sin^2(\alpha) + \cos^2(\alpha) = 1$$

$$(-0.5)^2 + \cos^2(\alpha) = 1$$

$$\cos^2(\alpha) = 0.75$$

$$\cos(\alpha) = \pm\sqrt{0.75}$$

Of these, only $\cos(\alpha) = \sqrt{0.75}$ is valid because $\cos(\alpha)$ can only take positive values in the interval $\left[\frac{3\pi}{2}, \frac{5\pi}{2}\right]$.

Exercise

Solve the following equations:

(a) $\sin(2x) = 2 \cos(x)$

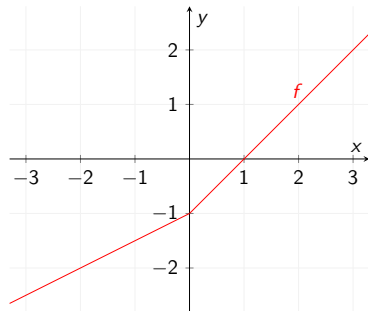
(b) $\cos^2(x) + \sin(4x - 3) + \sin^2(x) = \frac{\sqrt{3}+2}{2}$

(c) $2 \cos^2(x) + \cos(x) = 1$

(d) $\sin(2x) = \sin(x)$.

Piecewise Defined Functions

The function f in the image below clearly consists of two (half)lines.



It cannot be expressed with a single formula, but it can be written in the form

$$f(x) = \begin{cases} \frac{x}{2} - 1, & x < 0 \\ x - 1, & x \geq 0 \end{cases}.$$

This notation means that the function f follows the expression $\frac{x}{2} - 1$ for negative values of the variable and the expression $x - 1$ for non-negative values. Functions defined this way, with different expressions for different ranges of the variable, are called **piecewise defined functions**.

Since the function values are unique, in a piecewise defined function, each value of the variable must satisfy only one condition. In the previous function, the case $x = 0$ is included in the lower condition, but it could just as well have been included in the upper condition since both expressions yield the same value at this point. However, for example, in the case of the function

$$f(x) = \begin{cases} \frac{x}{2} - 1, & x < 0 \\ x + 1, & x \geq 0 \end{cases}$$

the case $x = 0$ must be assigned to the expression that gives the desired value of the function at $x = 0$.

A piecewise defined function can consist of more than two expressions, as in the function

$$f(x) = \begin{cases} \cos(x), & x < 1 \\ 1, & x = 1 \\ \ln(x), & x \geq 1 \end{cases}.$$

Example

The speed of an elevator is 1 m/s. The elevator moves upwards for 15 seconds, then stops for 5 seconds, and then moves downwards for 9 seconds. If accelerations are neglected, the motion of the elevator can be described by the function

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 15 \\ 0, & 15 < x \leq 20 \\ -1, & 20 < x \leq 29 \end{cases}$$

Sketch the graph of the function.

In this case, changing the equality conditions slightly alters the function. Why?

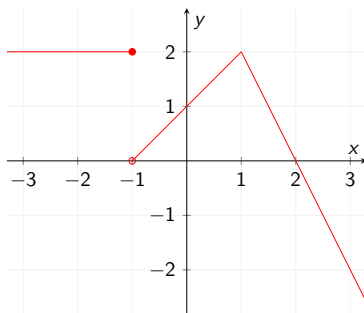
Basic Exercise

(a) Consider the function

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = \begin{cases} e^{-x}, & x < 0 \\ x^2, & \text{otherwise} \end{cases}.$$

Calculate $f(-1)$, $f(0)$, and $f(2)$.

(b) Which function is represented in the following graph?



(c) Draw the graph of the function from part (a).

If an equation involves a piecewise defined function, there are actually two or more equations hidden within it. For example, in the case of the function

$$f(x) = \begin{cases} -x + 1, & x < 0 \\ e^x, & x \geq 0 \end{cases}$$

the equation $f(x) = 2$ means that for $x < 0$ the equation is

$$-x + 1 = 2 \iff x = -1$$

and for $x \geq 0$ the equation is

$$e^x = 2 \iff x = \ln(2).$$

Note that the conditions $x < 0$ and $x \geq 0$ are indeed significant for the solution. In the previous solution, the candidate solutions satisfied their respective conditions, so both are indeed solutions.

However, in the case

$$f(x) = \begin{cases} x + 1, & x < 0 \\ e^x, & x \geq 0 \end{cases}$$

the situation is different: When $x < 0$, we have

$$2 = x + 1 \iff x = 1,$$

but since this candidate does not satisfy the condition $x < 0$, it is not a solution to the equation. This can be confirmed by substituting the solution back into the original equation, which results in $2 = e$, or by solving the equation graphically (Solve it!). The other candidate remains a solution, which can be verified similarly by substitution or graphing.

Basic Exercise

Solve both algebraically and graphically the following equations:

$$(a) \quad f(x) = 2x - 1, \quad f(x) = \begin{cases} x + 3, & x < 0 \\ -x + 2, & x \geq 0 \end{cases}$$

$$(b) \quad f(x) = 4, \quad f(x) = \begin{cases} x^2, & x < \frac{1}{2} \\ \sin(x), & x \geq \frac{1}{2} \end{cases}.$$

Absolute Value

The definition of absolute value is

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases} . \quad (9)$$

If a more complex expression than the number x is inside the absolute value, the corresponding formula is

$$|f(x)| = \begin{cases} f(x), & f(x) \geq 0 \\ -f(x), & f(x) < 0 \end{cases} .$$

Example

Expand the absolute values in the expression $|(x + 3)(x - 4)|$.

Solution

$$\begin{aligned} |(x + 3)(x - 4)| &= \begin{cases} (x + 3)(x - 4), & (x + 3)(x - 4) \geq 0 \\ -(x + 3)(x - 4), & (x + 3)(x - 4) < 0 \end{cases} \\ &= \begin{cases} (x + 3)(x - 4), & x \leq -3 \text{ or } x \geq 4 \\ -(x + 3)(x - 4), & -3 < x < 4 \end{cases}, \end{aligned}$$

where the inequality $(x + 3)(x - 4) \geq 0$ is solved using a general technique for solving inequalities.

When there are multiple absolute values, in the worst case, each one must be handled separately, increasing the number of cases to be considered, as observed in the following example.

Example

Expand the absolute values in the expression $|f(x)| + |g(x)|$.

Solution

$$\begin{aligned} |f(x)| + |g(x)| &= \begin{cases} f(x) + |g(x)|, & f(x) \geq 0 \\ -f(x) + |g(x)|, & f(x) < 0 \end{cases} \\ &= \begin{cases} f(x) + g(x), & f(x) \geq 0 \text{ and } g(x) \geq 0 \\ f(x) - g(x), & f(x) \geq 0 \text{ and } g(x) < 0 \\ -f(x) + g(x), & f(x) < 0 \text{ and } g(x) \geq 0 \\ -f(x) - g(x), & f(x) < 0 \text{ and } g(x) < 0 \end{cases} \end{aligned}$$

In practice, the absolute value makes a number positive. Geometrically, $|x|$ represents the distance of the number x from the origin[†]. The expressions $|x - y|$ and $|y - x|$ describe the distance between the numbers x and y . See Figure below.

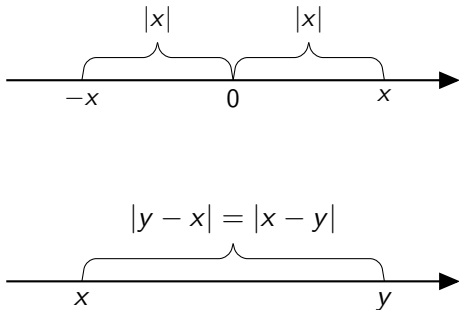


Figure: Absolute Value.

[†]This interpretation of absolute value is actually deeper than the aforementioned (making it positive), as it extends to complex numbers, which are a sort of two-dimensional numbers.

Based on the geometric interpretation of absolute value, the inequality $|x| < 1$ describes the numbers that are less than one unit away from the origin. For $a > 0$, it more generally holds that

$$|x| < a \iff -a < x < a \quad (10)$$

Similarly,

$$|x| > a \iff x > a \text{ or } x < -a.$$

(What does this mean geometrically?) Similarly, $|x - 2| < 1$ describes the numbers that are less than one unit away from the number 2. For $a > 0$, it more generally holds that

$$|x - b| < a \iff b - a < x < b + a. \quad (11)$$

Similarly,

$$|x - b| > a \iff x > b + a \text{ or } x < b - a.$$

These conditions actually work correctly even in the geometrically nonsensical cases where $a < 0$ or $a = 0$. This is why these properties can be used to solve inequalities of the form $|f(x)| < g(x)$.

Example

Solve the equation $|2x + 6| = 3x - 3$.

Solution

Method 1:

$$|2x + 6| = 3x - 3, \quad |2x + 6| = \begin{cases} 2x + 6, & x \geq -3 \\ -2x - 6, & x < -3 \end{cases}$$

For $x \geq -3$:

$$2x + 6 = 3x - 3$$

$$x = 9 \quad (\geq -3, \text{ is a solution})$$

For $x < -3$:

$$-2x - 6 = 3x - 3$$

$$5x = -3$$

$$x = -\frac{3}{5} \quad (\geq -3, \text{ not a solution})$$

Answer: $x = 9$.

Solution (continued)

Method 2:

$$2x + 6 = \pm(3x - 3)$$

$$2x + 6 = 3x - 3$$

$$x = 9$$

$$2x + 6 = -(3x - 3)$$

$$2x + 6 = -3x + 3$$

$$5x = -3$$

$$x = -\frac{3}{5} \text{ (not a solution as } -\frac{3}{5} < -3)$$

Answer: $x = 9$.

Example

Solve the equation $|x| = x$.

Solution

$$|x| = x$$

$$x = x \quad \text{or} \quad -x = x$$

$$0 = 0 \quad \text{or} \quad x = 0.$$

The latter equation thus gives one potential solution, while the former is true for all x . Checking the solution $x = 0$ is straightforward, but not for solutions of the first equation. It is clearest to determine for which values of the variable the equations hold.

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases},$$

so the original equation becomes $x = x$ when $x \geq 0$, and since this equation is always true, the original equation holds for $x \geq 0$. On the other hand, the original equation becomes $-x = x$ when $x < 0$. The solution $x = 0$ does not satisfy the condition $x < 0$. Therefore, $x = 0$ is not a solution based on the latter equation but only from the first equation. Thus, the overall solution is $x \geq 0$.

Example

Solve the equation $|x^2 - 1| = |2x + 2|$.

Solution

Method 1:

$$|x^2 - 1| = \begin{cases} x^2 - 1, & x \leq -1 \text{ or } x \geq 1 \\ -x^2 + 1, & -1 < x < 1 \end{cases}, \quad |2x + 2| = \begin{cases} 2x + 2, & x \geq -1 \\ -2x - 2, & x < -1 \end{cases}.$$

Three cases: $x < -1$, $-1 < x < 1$, and $x \geq 1$ or $x = -1$.

$x < -1$:

$$x^2 - 1 = -2x - 2$$

$$x^2 + 2x + 1 = 0$$

$$x = -1 \quad (\text{not a solution})$$

Solution (continued)

$$-1 < x < 1:$$

$$-x^2 + 1 = 2x + 2$$

$$x^2 + 2x + 1 = 0$$

$$x = -1 \quad (\text{not a solution})$$

$$x \geq 1 \text{ or } x = -1:$$

$$x^2 - 1 = 2x + 2$$

$$x^2 - 2x - 3 = 0$$

$$x = -1 \quad \text{or} \quad x = 3$$

Answer: $x = -1$ or $x = 3$.

Solution (continued)

The equation was initially written out separately on both sides. If you want to avoid this, you should proceed as in Example 1.133:

$$\begin{aligned} 0 &= |2x + 2| - |x^2 - 1| \\ &= \begin{cases} 2x + 2 - |x^2 - 1|, & x \geq -1 \\ -2x - 2 - |x^2 - 1|, & x < -1 \end{cases} \\ &= \begin{cases} 2x + 2 - (x^2 - 1), & x \geq -1 \text{ and } (x \leq -1 \text{ or } x \geq 1) \\ 2x + 2 + (x^2 - 1), & x \geq -1 \text{ and } -1 < x < 1 \\ -2x - 2 - (x^2 - 1), & x < -1 \text{ and } (x \leq -1 \text{ or } x \geq 1) \\ -2x - 2 + (x^2 - 1), & x < -1 \text{ and } -1 < x < 1 \end{cases} \\ &= \begin{cases} -x^2 + 2x + 3, & x = -1 \text{ or } \geq 1 \\ x^2 + 2x + 1, & -1 < x < 1 \\ -x^2 - 2x - 1, & x < -1 \\ x^2 - 2x - 3, & \text{never} \end{cases} = \begin{cases} -x^2 + 2x + 3, & x = -1 \text{ or } \geq 1 \\ x^2 + 2x + 1, & -1 < x < 1 \\ -x^2 - 2x - 1, & x < -1 \end{cases} \end{aligned}$$

For this monstrous expression, it is good to check that each value of the variable corresponds to only one expression.

Solution (continued)

Method 2: Possible cases:

$$x^2 - 1 = \pm(2x + 2) \quad \text{and} \quad -x^2 + 1 = \pm(2x + 2).$$

Different cases:

$$x^2 - 1 = \pm(2x + 2).$$

$$x^2 - 1 = 2x + 2$$

$$x^2 - 2x - 3 = 0$$

$$x = -1 \quad \text{or} \quad x = 3$$

$$x^2 - 1 = -2x - 2$$

$$x^2 + 2x + 1 = 0$$

$$x = -1,$$

where both satisfy the original equation.

Answer: $x = -1$ or $x = 3$.

Basic Exercise

Solve the following equations and inequalities:

$$(a) |3x + 7| = 4 \quad (b) |5x - 2| = -3 \quad (c) |3x + 7| < 4 \quad (d) |2x - 1| = x - 2.$$

Part (c) of the previous task can be solved using the general technique for solving inequalities, by expanding the inequality based on the definition, or using the property (10).

Example

Solve the inequality $|x + 1| < -2x + 3$.

Solution

Use the general technique for solving inequalities. Determine when the function $f(x) = |x + 1| + 2x - 3$ is negative.

There are no discontinuities.

Solution (continued)

Zeros:

$$f(x) = 0$$

$$|x + 1| + 2x - 3 = 0$$

$$|x + 1| = -2x + 3$$

$$x + 1 = -2x + 3 \quad \text{or} \quad -x - 1 = -2x + 3$$

$$x = \frac{2}{3} \quad \text{or} \quad x = 4,$$

where only $x = \frac{2}{3}$ is actually a zero (Why isn't the other one?).

Sign of the function f on the intervals $(-\infty, \frac{2}{3})$ and $(\frac{2}{3}, \infty)$:

$$f(0) = |0 + 1| + 2 \cdot 0 - 3 = -2 < 0,$$

so $f(x) < 0$ on the interval $(-\infty, \frac{2}{3})$.

$$f(1) = |1 + 1| + 2 \cdot 1 - 3 = 1 > 0,$$

so $f(x) > 0$ on the interval $(\frac{2}{3}, \infty)$.

Answer: $x \in (-\infty, \frac{2}{3})$.

Exercise

Solve the following equations and inequalities:

(a) $|2x - 1| = |x - 2|$

(b) $||x| + 2| = 3$

(c) $|x + 2| + |2x| = 3$

(d) $|x + 2| + |2x| > 3.$

Since absolute values make negative numbers positive, taking the absolute value of a function means that the parts of the graph below the x -axis are reflected above the x -axis.

Basic Exercise

Draw the graphs of the following functions (without a calculator):

(a) $f(x) = |x|$

(b) $f(x) = |x| - 2$

(c) $f(x) = ||x| - 2| - 1$

(d) $f(x) = |x^2 - 1|$.

Lastly, let's list some rules for absolute values:

For all $a, b \in \mathbb{R}$, the following properties hold:

$$(1) |a| \geq 0$$

$$(2) |a| = 0 \iff a = 0$$

$$(3) \sqrt{a^2} = |a|$$

$$(4) |-a| = |a|$$

$$(5) |ab| = |a| |b|$$

$$(6) \left| \frac{a}{b} \right| = \frac{|a|}{|b|}, b \neq 0$$

$$(7) |a^n| = |a|^n, n \in \mathbb{Z}$$

$$(8) |a + b| \leq |a| + |b|.$$

Note immediately: The formula $|a + b| = |a| + |b|$ does not hold for absolute values, but only the inequality in item 8, called the **triangle inequality**.

Basic Exercise

Provide a counterexample to show that $|a + b| \neq |a| + |b|$ for absolute values.

The remaining listed properties are likely easy to accept. Based on these, one can simplify expressions such as:

$$|-3x| = |-3| |x| = 3 |x|.$$

Note also that item (7) also holds for all real exponents n . If n is even, it also holds that $|a^n| = |a|^n = a^n$.

Example

The incorrectly solved inequality $x^2 - 4 > 0$ in the introduction can be solved using properties of absolute values as follows:

$$x^2 - 4 > 0$$

$$x^2 > 4$$

$$\sqrt{x^2} > \sqrt{4}$$

$$|x| > 2$$

$$-2 < x < 2.$$

The second and third inequalities are equivalent based on the fact that the square root function is strictly increasing. Also note that no negative numbers appear under the square root, even though the reasoning holds for negative values of the variable x as well.

Counterexamples

If someone claims that every person has dark hair, how would you prove them wrong? Exactly – you show a person whose hair is not dark. This works because the property was claimed to apply to **all** people, so a single counterexample is enough to prove the statement false. On the other hand, if the claim were that most people have dark hair, then one or even several counterexamples would not be enough to refute the claim. The principle of counterexamples applies generally, regardless of the subject of the statement.

If a statement concerns all cases of a certain kind, then to prove the statement false, it suffices to find just one case in which the statement is not true. Such a case is called a **counterexample**.

Counterexamples are extremely useful in mathematics because mathematical statements usually concern all cases of a certain kind. For instance, if we are considering whether

$$\cos(2x) = 2 \cos(x)$$

holds, we can quickly investigate this using a counterexample. If there is even one value of the variable x for which the equation is not true, then the formula does not hold. Now, for example, in the case $x = 0$, we have $\cos(2 \cdot 0) = \cos(0) = 1$ and $2 \cos(0) = 2 \cdot 1 = 2$, so the formula is not true in this case, and hence $\cos(2x) \neq 2 \cos(x)$.

Earlier, it was noted that mathematical statements often concern all cases of a certain kind. This matter needs to be addressed a bit more precisely, as often, especially in the case of formulas, there is some imprecision about this. For example, the formula

$$\sin(2x) = 2 \sin(x) \cos(x)$$

means that the given equation holds for **every** $x \in \mathbb{R}$, even though this is not explicitly stated but assumed to be clear from the context. Similarly, the formula

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

holds for every $n \in \mathbb{Z}_+$, even if this is not explicitly stated. It would thus be more precise to write, for example,

$$\sin(2x) = 2 \sin(x) \cos(x), \quad x \in \mathbb{R},$$

and

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}, \quad n \in \mathbb{Z}_+,$$

but sometimes we are a bit lazy or strive for concise text. When we previously pondered whether $\cos(2x) = 2 \cos(x)$ holds, what was actually meant was whether the given equation holds for **every** $x \in \mathbb{R}$. Thus, one counterexample was sufficient to disprove the statement.

If, when looking for a counterexample, the statement happens to work on the first try, this does not mean that the statement is true, but that then more counterexamples need to be tried.

For example, the equation $\sin(2x) = 2\sin(x)$ happens to hold in the case $x = 0$, but in the case $x = \frac{\pi}{2}$, for example, the equation does not hold, so the formula is not true. Similarly, the failure of even several counterexamples does not prove the formula to be true. For example, the equation $\sin(2x) = 2\sin(x)$ holds in every case $x = 2\pi n$, $n \in \mathbb{Z}$, and although there are infinitely many such cases, the formula is not true because cases can be found where it is not true.

If no counterexample seems to be found, the statement **may** be true, but this still needs to be proven in some way to gain certainty. For example, the equation $\sin^2(x) + \cos^2(x) = 1$ is true, but even investigating a million individual cases is not enough to prove it, because there are still infinitely many cases left to consider, some of which may not satisfy the equation.

Individual cases can often easily show a statement to be false, but usually, they cannot be used to show a statement to be true.

Let's write an example of how proving a statement to be false could be neatly formulated.

Example

Show that $\cos(2x) = 2 \cos(x)$ does not hold.

Solution

Let's prove the formula $\cos(2x) = 2 \cos(x)$ false by using a counterexample.

Counterexample. In the case $x = 0$, we have $\cos(2 \cdot 0) = \cos(0) = 1$ and $2 \cos(0) = 2 \cdot 1 = 2$, so the equation is not true in this case. Thus, the examined formula is false.

Basic Exercise

Use counterexamples to determine which of the following formulas do not hold:

(a) $(a + b)^2 = a^2 + b^2$

(b) $e^{a+b} = e^a + e^b$

(c) $\cos(a + b) = \cos(a) + \cos(b)$

(d) $\sin(a + b) = \sin(a) + \sin(b)$

(e) $|a + b| = |a| + |b|$.

Exercise

Show the statement “Every polynomial of degree n has at most n real roots” to be false.

In some cases, finding one or more counterexamples does not make the statement entirely useless. For example, the statement considered in the previous task can be corrected to be true by limiting it to non-zero polynomials. The statement “Every non-zero polynomial of degree n has at most n real roots” is thus true.

Note also that the previous statement did not concern any formula. Many mathematical propositions are like this, and the counterexample principle applies to them in the same way as it does to formulas, as long as the proposition concerns all cases of a certain kind. This is usually the case. Exceptions mainly include propositions that state that there exists (i.e., at least one) a certain kind of mathematical object, such as a certain function value, etc. (Compare: There exists a person with red hair.) Such propositions cannot be disproven with a counterexample.

Let's also mention as a contrasting example a case, where restricting the situation to only the cases that work is not reasonable. The formula $|a + b| = |a| + |b|$ does **NOT** hold because it is not true when the numbers a and b have opposite signs. If these cases are ruled out (like the zero polynomial was ruled out above), the formula is true, but it is not useful because when the numbers are positive, it is also true that $|a + b| = a + b$ and when the numbers are negative, $|a + b| = -a - b$, which are more useful simplifications. On the other hand, the formula would be highly misleading because it is defined[†] in every case, but it works in only some of them.

Cases where the statement is not defined are not counted as counterexamples. For example, the equation

$$\tan(2x) = \frac{2 \tan(x)}{1 - \tan^2(x)}$$

is not defined in cases $x = \frac{\pi}{2} + \pi n$ and $x = \frac{\pi}{4} + \frac{\pi}{2}n$, $n \in \mathbb{Z}$, so it cannot be true in these cases. However, since the equation holds in all cases where it is defined, it is a useful formula.

[†]That is, both sides of it can be calculated for all possible values of the variables.

Basic Exercise

Consider the following formulas:

$$(i) \sqrt{a+b} = \sqrt{a} + \sqrt{b}$$

$$(ii) \ln(a+b) = \ln(a) + \ln(b)$$

$$(iii) \tan(a+b) = \tan(a) + \tan(b)$$

$$(iv) \frac{1}{a+b} = \frac{1}{a} + \frac{1}{b}.$$

- (a) For which values of the real numbers a and b are the formulas defined?
- (b) Using counterexamples, determine which formulas are not true.

Basic Exercise

Show that the formula $\tan(2x) = 2 \tan(x)$ does not hold.

Exercise

Consider the statement “The square of every real number is positive.”

- (a) Show that the statement is false.
- (b) Modify the statement so that it is true.

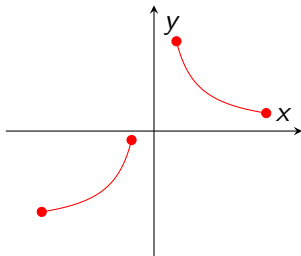
A valid picture can also serve as a counterexample, as long as it is appropriate. It should also be noted that a picture alone rarely suffices to prove a statement.

Example

The statement

"If a function does not take the same value twice, then it is either strictly increasing or strictly decreasing"

is not true. The following picture is sufficient as a counterexample, as it clearly shows a function that does not take the same value twice but is not strictly increasing or strictly decreasing.



The Significance of Counterexamples

- Instead of using counterexamples, a claim can also be shown to be false by demonstrating how the situation actually works. However, this is often laborious, and **the main significance of counterexamples is that they are almost always the simplest, clearest, and quickest way to show a claim is false.** For example, a counterexample can be found in just a few seconds, whereas proving how the situation actually works could easily take dozens of hours.
- Understanding the principle of counterexamples precisely is useful even for general education. For example, it is important to understand that if a claim does not concern all cases, then a single counterexample is not enough to refute it. On the other hand, claims concerning the average case cannot be shown to be false with a counterexample either. In everyday language, such claims are often phrased imprecisely by omitting the word referring to the average, assuming the context makes it clear. For example, if it is claimed that men are taller than women, a single counterexample of a woman being taller than a man is not sufficient to prove the claim false because the claim means that the average height of men is greater than that of women.

- In mathematics learning materials, counterexamples are usually used to show that a statement does not hold in the reverse direction. For example, when it is mentioned in learning materials that if a function is strictly increasing or decreasing, then it is an injection (meaning it does not take the same value twice, i.e., its graph does not reach the same height twice), it is often added that the statement does not hold in the reverse direction, with a counterexample provided as justification. A suitable counterexample for this situation was seen in an earlier example.
- Counterexamples are often used to justify why typically recurring mistakes are indeed mistakes. For instance, a common mistake is to write

$$\ln(a + b) = \ln(a) + \ln(b).$$

This does **NOT** hold, which can be justified to oneself with a counterexample.

- Counterexamples are also useful in inventing new mathematics, as they are a convenient way to test the validity of new ideas. When pondering whether something could work in a certain way, it is advisable to first try to refute the idea with counterexamples. If a counterexample does not seem to be found, the idea appears to be reasonable and worth attempting to prove.